DISCO-TAB — Comprehensive Benchmark

all generators × 24 datasets × all metrics; cells are **mean ± 95% confidence interval** over random seeds (Student-*t*, up to 3 seeds); **bold** = best per row (by mean, where a direction is defined); – = missing run/metric; ± absent where only one seed is available.

Utility (downstream TSTR)

Table 1: TSTR F1-macro (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.669 ± 0.487	0.483 ± 0.237	0.135 ± 0.582	0.591 ± 0.252	0.406	0.635 ± 0.151	0.396 ± 0.174	0.600 ± 0.270	0.559 ± 0.371	0.394 ± 0.217	0.388	**0.795**	0.787	0.778
Breast Cancer	0.922 ± 0.031	0.295 ± 0.304	0.934 ± 0.084	0.912 ± 0.041	0.924	0.934 ± 0.044	0.497 ± 0.461	**0.937 ± 0.047**	0.855 ± 0.082	0.513 ± 0.508	–	–	0.892	–
Liver Disorders	0.541 ± 0.168	0.473 ± 0.224	0.527 ± 0.171	0.553 ± 0.098	**0.592**	0.542 ± 0.174	0.488 ± 0.263	0.490 ± 0.333	0.500 ± 0.106	0.380 ± 0.208	0.522	–	0.520	–
Parkinsons	0.707 ± 0.300	0.480 ± 0.344	0.000 ± 0.000	0.630 ± 0.104	**0.928**	0.737 ± 0.215	0.424 ± 0.012	0.704 ± 0.131	0.695 ± 0.096	0.446 ± 0.139	–	–	0.400	–
Obesity	**0.861 ± 0.044**	0.138 ± 0.161	0.490 ± 0.086	0.243 ± 0.045	0.795	0.655 ± 0.097	0.115 ± 0.033	0.607 ± 0.092	0.171 ± 0.091	0.102 ± 0.026	–	–	0.637	–
Diabetes	0.842 ± 0.097	0.836 ± 0.042	0.875 ± 0.003	0.485 ± 0.014	**0.892 ± 0.006**	0.797 ± 0.086	0.478 ± 0.000	0.727 ± 0.121	0.658 ± 0.070	0.377 ± 0.082	0.517	–	0.862	0.557
German Credit	0.515 ± 0.581	0.448 ± 0.046	0.137 ± 0.591	0.496 ± 0.118	0.454 ± 0.107	0.594 ± 0.194	0.468 ± 0.203	0.501 ± 0.164	0.449 ± 0.145	0.459 ± 0.077	–	–	**0.657**	–
Bank Marketing	0.643 ± 0.302	0.604 ± 0.039	**0.673 ± 0.043**	0.469 ± 0.001	0.637 ± 0.018	0.608 ± 0.082	0.469 ± 0.000	0.498 ± 0.058	0.471 ± 0.005	0.236 ± 0.317	0.482	–	0.609	–
Census Income	0.748 ± 0.048	0.727 ± 0.031	0.746 ± 0.014	0.459 ± 0.038	0.750 ± 0.020	0.669 ± 0.126	0.433 ± 0.002	0.549 ± 0.126	0.615 ± 0.145	0.524 ± 0.264	0.662	–	**0.769**	–
Melanoma	0.576 ± 0.125	0.406 ± 0.142	0.000 ± 0.000	0.606 ± 0.185	0.510 ± 0.063	0.557 ± 0.099	0.442 ± 0.070	0.529 ± 0.128	0.508 ± 0.201	0.445 ± 0.588	0.584	–	**0.644**	–
PBC	0.472 ± 0.433	0.490 ± 0.120	0.376 ± 0.000	0.534 ± 0.108	0.564 ± 0.019	0.590 ± 0.152	0.492 ± 0.182	**0.646 ± 0.177**	0.566 ± 0.220	0.428 ± 0.123	0.469	–	0.635	–
PBC3	0.607 ± 0.378	0.475 ± 0.105	0.000 ± 0.000	0.530 ± 0.014	0.453 ± 0.000	**0.632 ± 0.095**	0.453 ± 0.000	0.469 ± 0.098	0.477 ± 0.101	0.497 ± 0.131	0.453	–	0.617	0.623
Retinopathy	0.538 ± 0.053	0.509 ± 0.257	0.124 ± 0.535	0.484 ± 0.040	0.452 ± 0.101	0.539 ± 0.211	0.480 ± 0.201	0.478 ± 0.147	0.513 ± 0.146	0.511 ± 0.042	**0.562**	–	0.533	–
WHAS500	0.713 ± 0.102	0.428 ± 0.040	0.476 ± 0.227	0.670 ± 0.213	**0.763 ± 0.059**	0.699 ± 0.159	0.395 ± 0.161	0.619 ± 0.114	0.709 ± 0.046	0.324 ± 0.096	–	–	0.643	0.637
GBSG2	0.575 ± 0.039	0.438 ± 0.073	0.505 ± 0.182	0.526 ± 0.181	0.547 ± 0.135	0.538 ± 0.086	0.474 ± 0.204	0.556 ± 0.082	0.590 ± 0.196	0.475 ± 0.310	0.460	–	**0.617**	–
Colon	0.624 ± 0.081	0.506 ± 0.146	0.534 ± 0.031	0.541 ± 0.033	0.577 ± 0.063	0.497 ± 0.250	0.426 ± 0.203	0.564 ± 0.075	0.526 ± 0.083	0.450 ± 0.097	0.548	–	**0.659**	–
GRACE	**0.615 ± 0.035**	0.456 ± 0.103	0.437 ± 0.083	0.610 ± 0.067	0.576 ± 0.068	0.591 ± 0.100	0.456 ± 0.074	0.508 ± 0.058	0.543 ± 0.224	0.463 ± 0.033	–	–	0.607	–
TRACE	0.642 ± 0.082	0.513 ± 0.077	0.666 ± 0.081	0.624 ± 0.028	0.660 ± 0.121	0.646 ± 0.163	0.445 ± 0.065	0.516 ± 0.148	0.609 ± 0.192	0.516 ± 0.045	–	–	**0.710**	–
Framingham	0.569 ± 0.096	**0.577 ± 0.090**	0.455 ± 0.046	0.488 ± 0.036	0.556 ± 0.031	0.546 ± 0.088	0.474 ± 0.112	0.486 ± 0.104	0.453 ± 0.122	0.448 ± 0.110	–	–	0.512	–
Blood Transfusion	0.330 ± 0.711	0.481 ± 0.151	0.000 ± 0.000	0.579 ± 0.146	0.541 ± 0.065	0.498 ± 0.141	0.510 ± 0.243	0.517 ± 0.078	0.456 ± 0.215	0.502 ± 0.198	–	–	**0.624**	–
QSAR Biodeg	0.724 ± 0.098	0.419 ± 0.028	0.662 ± 0.126	0.712 ± 0.106	0.391 ± 0.025	0.729 ± 0.081	0.428 ± 0.232	**0.770 ± 0.071**	0.612 ± 0.298	0.393 ± 0.338	–	–	0.651	–
Maternal Risk	**0.615 ± 0.199**	0.305 ± 0.042	0.000 ± 0.000	0.375 ± 0.160	0.296 ± 0.087	0.463 ± 0.240	0.235 ± 0.187	0.527 ± 0.099	0.395 ± 0.082	0.234 ± 0.260	0.457	–	0.549	0.476
Website Phishing	**0.614 ± 0.075**	0.329 ± 0.130	0.521 ± 0.049	0.359 ± 0.081	0.604 ± 0.064	0.555 ± 0.090	0.263 ± 0.217	0.492 ± 0.047	0.454 ± 0.013	0.286 ± 0.087	–	–	0.534	–

Table 2: TSTR F1-weighted (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.710 ± 0.406	0.559 ± 0.243	0.185 ± 0.796	0.679 ± 0.178	0.555	0.710 ± 0.122	0.500 ± 0.317	0.676 ± 0.173	0.655 ± 0.256	0.416 ± 0.438	0.530	**0.827**	0.823	0.812
Breast Cancer	0.927 ± 0.031	0.284 ± 0.354	0.939 ± 0.079	0.920 ± 0.038	0.929	0.939 ± 0.042	0.489 ± 0.601	**0.942 ± 0.045**	0.869 ± 0.068	0.558 ± 0.428	–	–	0.901	–
Liver Disorders	0.543 ± 0.165	0.472 ± 0.228	0.529 ± 0.169	0.553 ± 0.099	**0.592**	0.543 ± 0.175	0.488 ± 0.265	0.490 ± 0.333	0.501 ± 0.105	0.384 ± 0.206	0.522	–	0.521	–
Parkinsons	0.761 ± 0.283	0.632 ± 0.164	0.000 ± 0.000	0.729 ± 0.116	**0.947**	0.801 ± 0.156	0.630 ± 0.018	0.784 ± 0.111	0.768 ± 0.100	0.584 ± 0.126	–	–	0.595	–
Obesity	**0.863 ± 0.046**	0.138 ± 0.168	0.493 ± 0.088	0.250 ± 0.046	0.796	0.655 ± 0.108	0.120 ± 0.036	0.610 ± 0.098	0.176 ± 0.096	0.101 ± 0.023	–	–	0.636	–
Diabetes	0.951 ± 0.033	0.952 ± 0.017	0.962 ± 0.001	0.876 ± 0.003	**0.968 ± 0.002**	0.945 ± 0.021	0.874 ± 0.001	0.927 ± 0.029	0.901 ± 0.039	0.584 ± 0.144	0.881	–	0.961	0.751
German Credit	0.500 ± 0.735	0.585 ± 0.036	0.192 ± 0.827	0.624 ± 0.074	0.604 ± 0.067	0.680 ± 0.112	0.601 ± 0.099	0.631 ± 0.105	0.587 ± 0.049	0.516 ± 0.219	–	–	**0.730**	–
Bank Marketing	0.862 ± 0.061	0.846 ± 0.049	0.854 ± 0.030	0.828 ± 0.000	**0.872 ± 0.004**	0.862 ± 0.019	0.828 ± 0.000	0.835 ± 0.015	0.829 ± 0.001	0.273 ± 0.618	0.831	–	0.865	–
Census Income	0.819 ± 0.020	0.812 ± 0.018	0.818 ± 0.016	0.671 ± 0.019	0.826 ± 0.013	0.782 ± 0.067	0.656 ± 0.002	0.718 ± 0.069	0.748 ± 0.076	0.672 ± 0.164	0.776	–	**0.833**	–
Melanoma	0.611 ± 0.164	0.573 ± 0.124	0.000 ± 0.000	0.706 ± 0.136	0.660 ± 0.041	0.676 ± 0.045	0.588 ± 0.059	0.645 ± 0.130	0.614 ± 0.026	0.454 ± 0.688	0.678	–	**0.737**	–
PBC	0.449 ± 0.518	0.502 ± 0.119	0.454 ± 0.000	0.544 ± 0.102	0.584 ± 0.026	0.592 ± 0.167	0.528 ± 0.205	**0.662 ± 0.191**	0.586 ± 0.156	0.444 ± 0.124	0.492	–	0.651	–
PBC3	0.699 ± 0.461	0.756 ± 0.039	0.000 ± 0.000	0.780 ± 0.013	0.751 ± 0.000	0.786 ± 0.079	0.751 ± 0.000	0.749 ± 0.037	0.748 ± 0.012	0.675 ± 0.082	0.751	–	**0.804**	0.712
Retinopathy	0.558 ± 0.058	0.526 ± 0.279	0.148 ± 0.636	0.515 ± 0.040	0.480 ± 0.099	0.565 ± 0.183	0.481 ± 0.265	0.507 ± 0.136	0.531 ± 0.148	0.517 ± 0.032	**0.571**	–	0.557	–
WHAS500	0.727 ± 0.083	0.493 ± 0.047	0.538 ± 0.159	0.694 ± 0.198	**0.779 ± 0.055**	0.719 ± 0.127	0.428 ± 0.101	0.642 ± 0.126	0.721 ± 0.042	0.339 ± 0.098	–	–	0.668	0.656
GBSG2	0.582 ± 0.031	0.461 ± 0.045	0.530 ± 0.145	0.539 ± 0.168	0.558 ± 0.123	0.542 ± 0.110	0.486 ± 0.213	0.564 ± 0.073	0.596 ± 0.186	0.474 ± 0.344	0.477	–	**0.631**	–
Colon	0.623 ± 0.082	0.506 ± 0.148	0.534 ± 0.033	0.541 ± 0.032	0.577 ± 0.062	0.495 ± 0.255	0.429 ± 0.200	0.565 ± 0.074	0.526 ± 0.084	0.451 ± 0.092	0.550	–	**0.659**	–
GRACE	0.643 ± 0.038	0.551 ± 0.075	0.566 ± 0.057	0.668 ± 0.066	0.651 ± 0.047	0.656 ± 0.050	0.526 ± 0.150	0.591 ± 0.033	0.634 ± 0.153	0.503 ± 0.096	–	–	**0.670**	–
TRACE	0.643 ± 0.081	0.513 ± 0.075	0.665 ± 0.082	0.624 ± 0.029	0.660 ± 0.120	0.645 ± 0.166	0.444 ± 0.069	0.516 ± 0.147	0.609 ± 0.191	0.516 ± 0.047	–	–	**0.710**	–
Framingham	0.613 ± 0.123	**0.641 ± 0.084**	0.586 ± 0.025	0.606 ± 0.025	0.639 ± 0.022	0.635 ± 0.054	0.590 ± 0.057	0.580 ± 0.101	0.580 ± 0.057	0.490 ± 0.119	–	–	0.623	–
Blood Transfusion	0.406 ± 0.874	0.520 ± 0.232	0.000 ± 0.000	0.676 ± 0.122	0.647 ± 0.058	0.607 ± 0.048	0.636 ± 0.128	0.615 ± 0.105	0.590 ± 0.192	0.551 ± 0.266	–	–	**0.715**	–
QSAR Biodeg	0.745 ± 0.085	0.541 ± 0.016	0.709 ± 0.105	0.755 ± 0.090	0.519 ± 0.034	0.767 ± 0.066	0.465 ± 0.314	**0.799 ± 0.059**	0.674 ± 0.210	0.361 ± 0.442	–	–	0.702	–
Maternal Risk	**0.643 ± 0.243**	0.340 ± 0.080	0.000 ± 0.000	0.447 ± 0.165	0.386 ± 0.041	0.510 ± 0.333	0.306 ± 0.153	0.592 ± 0.068	0.485 ± 0.087	0.251 ± 0.394	0.525	–	0.622	0.519
Website Phishing	**0.746 ± 0.068**	0.419 ± 0.082	0.717 ± 0.068	0.497 ± 0.115	0.737 ± 0.041	0.694 ± 0.052	0.346 ± 0.299	0.662 ± 0.035	0.614 ± 0.058	0.368 ± 0.100	–	–	0.685	–

Table 3: TSTR accuracy (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.711 ± 0.394	0.572 ± 0.305	0.228 ± 0.980	0.733 ± 0.110	0.683	0.750 ± 0.110	0.572 ± 0.514	0.711 ± 0.086	0.717 ± 0.143	0.472 ± 0.335	0.633	**0.833**	**0.833**	0.817
Breast Cancer	0.927 ± 0.033	0.316 ± 0.278	0.939 ± 0.079	0.921 ± 0.038	0.930	0.939 ± 0.044	0.561 ± 0.437	**0.942 ± 0.045**	0.874 ± 0.055	0.579 ± 0.363	–	–	0.904	–
Liver Disorders	0.589 ± 0.055	0.502 ± 0.181	0.551 ± 0.144	0.556 ± 0.091	**0.594**	0.551 ± 0.191	0.493 ± 0.252	0.493 ± 0.330	0.507 ± 0.095	0.473 ± 0.162	0.522	–	0.522	–
Parkinsons	0.752 ± 0.321	0.675 ± 0.074	0.000 ± 0.000	0.744 ± 0.169	**0.949**	0.803 ± 0.147	0.735 ± 0.037	0.795 ± 0.127	0.769 ± 0.127	0.607 ± 0.224	–	–	0.667	–
Obesity	**0.864 ± 0.042**	0.154 ± 0.127	0.499 ± 0.077	0.297 ± 0.056	0.801	0.655 ± 0.120	0.149 ± 0.042	0.627 ± 0.083	0.270 ± 0.101	0.115 ± 0.024	–	–	0.643	–
Diabetes	0.952 ± 0.035	0.954 ± 0.022	0.963 ± 0.000	0.915 ± 0.001	**0.970 ± 0.002**	0.953 ± 0.015	0.915 ± 0.001	0.941 ± 0.020	0.909 ± 0.095	0.479 ± 0.151	0.916	–	0.965	0.680
German Credit	0.542 ± 0.502	0.663 ± 0.147	0.233 ± 1.004	0.700 ± 0.033	0.710 ± 0.022	0.712 ± 0.040	0.680 ± 0.045	0.715 ± 0.054	0.668 ± 0.094	0.545 ± 0.255	–	–	**0.755**	–
Bank Marketing	0.875 ± 0.044	0.858 ± 0.094	0.845 ± 0.072	0.883 ± 0.000	**0.896 ± 0.002**	0.889 ± 0.003	0.883 ± 0.000	0.884 ± 0.004	0.883 ± 0.000	0.266 ± 0.436	0.883	–	0.895	–
Census Income	0.823 ± 0.020	0.823 ± 0.016	0.821 ± 0.022	0.766 ± 0.005	**0.835 ± 0.012**	0.812 ± 0.032	0.758 ± 0.007	0.782 ± 0.034	0.784 ± 0.036	0.698 ± 0.183	0.803	–	0.834	–
Melanoma	0.593 ± 0.152	0.626 ± 0.152	0.000 ± 0.000	0.724 ± 0.126	0.724 ± 0.070	0.707 ± 0.000	0.626 ± 0.185	0.667 ± 0.195	0.634 ± 0.160	0.463 ± 0.655	0.683	–	**0.756**	–
PBC	0.519 ± 0.296	0.497 ± 0.120	0.603 ± 0.000	0.540 ± 0.104	0.587 ± 0.039	0.593 ± 0.159	0.561 ± 0.277	**0.667 ± 0.209**	0.603 ± 0.079	0.439 ± 0.127	0.492	–	0.651	–
PBC3	0.676 ± 0.517	0.824 ± 0.020	0.000 ± 0.000	**0.838 ± 0.020**	0.829 ± 0.000	0.781 ± 0.102	0.829 ± 0.000	0.810 ± 0.020	0.805 ± 0.102	0.652 ± 0.164	0.829	–	0.829	0.671

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Retinopathy	0.563 ± 0.070	0.532 ± 0.301	0.198 ± 0.853	0.541 ± 0.089	0.491 ± 0.103	$\mathbf{0.581 \pm 0.146}$	0.514 ± 0.220	0.527 ± 0.146	0.532 ± 0.155	0.514 ± 0.034	0.568	–	0.568	–
WHAS500	0.728 ± 0.067	0.573 ± 0.108	0.627 ± 0.015	0.706 ± 0.193	$\mathbf{0.785 \pm 0.053}$	0.728 ± 0.086	0.452 ± 0.080	0.649 ± 0.147	0.720 ± 0.046	0.333 ± 0.093	–	–	0.677	0.656
GBSG2	0.587 ± 0.031	0.505 ± 0.068	0.589 ± 0.063	0.548 ± 0.146	0.563 ± 0.105	0.560 ± 0.153	0.493 ± 0.230	0.565 ± 0.065	0.597 ± 0.180	0.488 ± 0.280	0.493	–	**0.645**	–
Colon	0.629 ± 0.077	0.508 ± 0.149	0.542 ± 0.014	0.542 ± 0.028	0.578 ± 0.065	0.550 ± 0.089	0.492 ± 0.193	0.569 ± 0.071	0.535 ± 0.066	0.475 ± 0.056	0.565	–	**0.661**	–
GRACE	0.632 ± 0.040	0.592 ± 0.050	0.680 ± 0.022	0.680 ± 0.078	0.685 ± 0.033	0.678 ± 0.029	0.545 ± 0.228	0.625 ± 0.062	$\mathbf{0.692 \pm 0.072}$	0.495 ± 0.097	–	–	0.690	–
TRACE	0.645 ± 0.079	0.516 ± 0.070	0.670 ± 0.069	0.625 ± 0.029	0.660 ± 0.120	0.660 ± 0.119	0.501 ± 0.159	0.518 ± 0.148	0.611 ± 0.187	0.519 ± 0.040	–	–	**0.710**	–
Framingham	0.605 ± 0.147	0.649 ± 0.099	0.683 ± 0.008	0.684 ± 0.015	0.667 ± 0.024	0.674 ± 0.046	0.660 ± 0.046	0.613 ± 0.132	0.668 ± 0.046	0.472 ± 0.125	–	–	**0.695**	–
Blood Transfusion	0.421 ± 0.911	0.511 ± 0.195	0.000 ± 0.000	0.695 ± 0.128	0.667 ± 0.067	0.645 ± 0.163	0.682 ± 0.061	0.623 ± 0.155	0.629 ± 0.228	0.542 ± 0.304	–	–	**0.738**	–
QSAR Biodeg	0.739 ± 0.085	0.660 ± 0.018	0.724 ± 0.094	0.771 ± 0.084	0.643 ± 0.068	0.779 ± 0.058	0.485 ± 0.296	$\mathbf{0.806 \pm 0.059}$	0.709 ± 0.106	0.423 ± 0.276	–	–	0.720	–
Maternal Risk	0.630 ± 0.273	0.341 ± 0.109	0.000 ± 0.000	0.487 ± 0.192	0.436 ± 0.140	0.524 ± 0.417	0.381 ± 0.316	0.626 ± 0.082	0.557 ± 0.057	0.289 ± 0.356	0.571	–	**0.681**	0.527
Website Phishing	$\mathbf{0.761 \pm 0.049}$	0.441 ± 0.062	0.749 ± 0.071	0.510 ± 0.129	0.749 ± 0.055	0.708 ± 0.079	0.434 ± 0.223	0.683 ± 0.030	0.637 ± 0.081	0.336 ± 0.109	–	–	0.703	–

Statistical fidelity

Table 4: Jensen–Shannon div. (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.042 ± 0.035	0.094 ± 0.020	0.200 ± 0.033	0.042 ± 0.002	0.194	0.070 ± 0.006	0.192 ± 0.132	0.265 ± 0.048	0.040 ± 0.004	0.189 ± 0.014	0.049	**0.013**	0.069	0.046
Breast Cancer	$\mathbf{0.013 \pm 0.014}$	0.143 ± 0.007	0.111 ± 0.022	0.027 ± 0.011	0.294	0.072 ± 0.019	0.568 ± 0.357	0.466 ± 0.024	0.024 ± 0.014	0.293 ± 0.032	–	–	0.210	–
Liver Disorders	0.031 ± 0.009	0.184 ± 0.038	0.129 ± 0.058	0.028 ± 0.005	**0.028**	0.113 ± 0.031	0.118 ± 0.095	0.450 ± 0.074	0.040 ± 0.004	0.316 ± 0.019	0.042	–	0.118	–
Parkinsons	$\mathbf{0.028 \pm 0.014}$	0.175 ± 0.029	0.136 ± 0.013	0.056 ± 0.013	0.227	0.120 ± 0.015	0.143 ± 0.175	0.578 ± 0.075	0.049 ± 0.004	0.269 ± 0.016	–	–	0.221	–
Obesity	$\mathbf{0.008 \pm 0.004}$	0.044 ± 0.017	0.090 ± 0.015	0.057 ± 0.011	0.044	0.047 ± 0.075	0.126 ± 0.168	0.158 ± 0.006	0.033 ± 0.011	0.200 ± 0.006	–	–	0.051	–
Diabetes	0.018 ± 0.006	0.030 ± 0.003	0.035 ± 0.014	0.067 ± 0.007	$\mathbf{0.002 \pm 0.001}$	0.062 ± 0.041	0.174 ± 0.103	0.094 ± 0.055	0.032 ± 0.020	0.225 ± 0.020	0.061	–	0.031	0.097
German Credit	0.202 ± 0.547	0.020 ± 0.008	0.171 ± 0.014	0.008 ± 0.003	$\mathbf{0.007 \pm 0.001}$	0.019 ± 0.001	0.046 ± 0.025	0.067 ± 0.009	0.023 ± 0.002	0.140 ± 0.009	–	–	0.018	–
Bank Marketing	0.012 ± 0.004	0.009 ± 0.005	0.040 ± 0.019	0.015 ± 0.006	$\mathbf{0.004 \pm 0.003}$	0.040 ± 0.016	0.088 ± 0.110	0.083 ± 0.034	0.040 ± 0.017	0.275 ± 0.032	0.051	–	0.015	–
Census Income	0.050 ± 0.050	0.019 ± 0.005	0.055 ± 0.022	0.049 ± 0.060	$\mathbf{0.007 \pm 0.003}$	0.082 ± 0.067	0.182 ± 0.344	0.145 ± 0.108	0.085 ± 0.029	0.347 ± 0.009	0.067	–	0.012	–
Melanoma	0.046 ± 0.083	0.118 ± 0.005	0.147 ± 0.076	0.038 ± 0.013	0.044 ± 0.022	0.077 ± 0.079	0.187 ± 0.239	0.331 ± 0.072	0.033 ± 0.005	0.127 ± 0.033	**0.020**	–	0.040	–
PBC	0.041 ± 0.075	0.022 ± 0.018	0.072 ± 0.011	$\mathbf{0.005 \pm 0.007}$	0.006 ± 0.002	0.021 ± 0.011	0.053 ± 0.012	0.093 ± 0.163	0.014 ± 0.009	0.070 ± 0.011	0.009	–	0.020	–
PBC3	0.039 ± 0.043	0.070 ± 0.005	0.141 ± 0.039	$\mathbf{0.022 \pm 0.010}$	0.155 ± 0.061	0.051 ± 0.009	0.182 ± 0.169	0.303 ± 0.043	0.031 ± 0.002	0.179 ± 0.015	0.041	–	0.056	0.076
Retinopathy	0.123 ± 0.365	0.019 ± 0.023	0.168 ± 0.018	0.009 ± 0.004	$\mathbf{0.006 \pm 0.005}$	0.025 ± 0.022	0.043 ± 0.044	0.092 ± 0.007	0.020 ± 0.010	0.032 ± 0.006	0.006	–	0.018	–
WHAS500	0.013 ± 0.022	0.075 ± 0.051	0.131 ± 0.032	$\mathbf{0.011 \pm 0.001}$	0.063 ± 0.029	0.030 ± 0.013	0.129 ± 0.116	0.244 ± 0.020	0.015 ± 0.005	0.129 ± 0.010	–	–	0.066	0.022
GBSG2	0.030 ± 0.077	0.062 ± 0.044	0.173 ± 0.026	$\mathbf{0.015 \pm 0.006}$	0.023 ± 0.008	0.081 ± 0.110	0.131 ± 0.218	0.163 ± 0.058	0.016 ± 0.003	0.258 ± 0.017	0.046	–	0.025	–
Colon	0.010 ± 0.013	0.020 ± 0.013	0.104 ± 0.051	$\mathbf{0.003 \pm 0.001}$	0.005 ± 0.002	0.051 ± 0.064	0.104 ± 0.082	0.087 ± 0.019	0.008 ± 0.004	0.132 ± 0.017	0.065	–	0.012	–
GRACE	0.011 ± 0.007	0.085 ± 0.052	0.120 ± 0.043	$\mathbf{0.008 \pm 0.001}$	0.013 ± 0.005	0.069 ± 0.042	0.225 ± 0.495	0.269 ± 0.055	0.015 ± 0.012	0.149 ± 0.019	–	–	0.031	–
TRACE	0.009 ± 0.012	0.055 ± 0.032	0.111 ± 0.078	0.009 ± 0.001	$\mathbf{0.003 \pm 0.002}$	0.058 ± 0.041	0.100 ± 0.047	0.180 ± 0.130	0.020 ± 0.009	0.134 ± 0.003	–	–	0.013	–
Framingham	0.017 ± 0.002	0.059 ± 0.052	0.095 ± 0.012	0.010 ± 0.003	$\mathbf{0.004 \pm 0.002}$	0.060 ± 0.020	0.297 ± 0.057	0.226 ± 0.165	0.035 ± 0.011	0.205 ± 0.008	–	–	0.029	–
Blood Transfusion	0.067 ± 0.130	0.135 ± 0.061	0.193 ± 0.072	0.037 ± 0.018	$\mathbf{0.029 \pm 0.004}$	0.173 ± 0.049	0.228 ± 0.092	0.425 ± 0.105	0.043 ± 0.020	0.333 ± 0.025	–	–	0.085	–
QSAR Biodeg	0.033 ± 0.003	0.065 ± 0.014	0.090 ± 0.010	0.094 ± 0.027	0.509 ± 0.001	0.039 ± 0.016	0.272 ± 0.226	0.144 ± 0.022	$\mathbf{0.029 \pm 0.005}$	0.391 ± 0.011	–	–	0.049	–
Maternal Risk	0.038 ± 0.017	0.289 ± 0.069	0.264 ± 0.025	0.249 ± 0.020	0.275 ± 0.119	0.235 ± 0.132	0.362 ± 0.175	0.476 ± 0.047	0.140 ± 0.020	0.360 ± 0.014	**0.025**	–	0.167	0.114
Website Phishing	0.100 ± 0.394	0.224 ± 0.001	0.259 ± 0.003	0.223 ± 0.000	$\mathbf{0.001 \pm 0.001}$	0.005 ± 0.006	0.060 ± 0.052	0.003 ± 0.005	0.008 ± 0.005	0.028 ± 0.004	–	–	0.003	–

Table 5: KS statistic (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.129 ± 0.109	0.201 ± 0.045	0.335 ± 0.035	0.084 ± 0.022	0.262	0.162 ± 0.021	0.356 ± 0.130	0.246 ± 0.036	0.101 ± 0.045	0.313 ± 0.032	0.117	**0.052**	0.130	0.129
Breast Cancer	0.089 ± 0.081	0.329 ± 0.017	0.252 ± 0.085	0.103 ± 0.016	0.348	0.191 ± 0.029	0.736 ± 0.296	0.432 ± 0.062	$\mathbf{0.088 \pm 0.040}$	0.489 ± 0.028	–	–	0.303	–
Liver Disorders	$\mathbf{0.055 \pm 0.039}$	0.415 ± 0.127	0.265 ± 0.099	0.074 ± 0.014	0.073	0.241 ± 0.021	0.278 ± 0.215	0.358 ± 0.098	0.123 ± 0.082	0.479 ± 0.063	0.130	–	0.292	–
Parkinsons	$\mathbf{0.092 \pm 0.105}$	0.382 ± 0.051	0.268 ± 0.028	0.098 ± 0.016	0.293	0.253 ± 0.069	0.301 ± 0.247	0.454 ± 0.050	0.112 ± 0.017	0.454 ± 0.037	–	–	0.279	–
Obesity	$\mathbf{0.068 \pm 0.023}$	0.239 ± 0.111	0.235 ± 0.033	0.216 ± 0.024	0.156	0.217 ± 0.063	0.354 ± 0.349	0.289 ± 0.023	0.161 ± 0.023	0.305 ± 0.017	–	–	0.193	–
Diabetes	0.064 ± 0.036	0.121 ± 0.025	0.097 ± 0.020	0.120 ± 0.042	$\mathbf{0.014 \pm 0.003}$	0.129 ± 0.030	0.262 ± 0.082	0.158 ± 0.058	0.130 ± 0.050	0.400 ± 0.028	0.116	–	0.090	0.233
German Credit	0.098 ± 0.025	0.125 ± 0.034	0.381 ± 0.075	0.046 ± 0.022	$\mathbf{0.039 \pm 0.002}$	0.099 ± 0.049	0.249 ± 0.094	0.143 ± 0.012	0.095 ± 0.041	0.359 ± 0.017	–	–	0.070	–
Bank Marketing	0.051 ± 0.013	0.126 ± 0.022	0.171 ± 0.025	0.212 ± 0.048	$\mathbf{0.020 \pm 0.007}$	0.169 ± 0.144	0.414 ± 0.263	0.285 ± 0.027	0.140 ± 0.018	0.643 ± 0.015	0.181	–	0.071	–
Census Income	0.080 ± 0.093	0.189 ± 0.135	0.227 ± 0.051	0.282 ± 0.273	$\mathbf{0.023 \pm 0.004}$	0.272 ± 0.021	0.492 ± 0.639	0.211 ± 0.127	0.224 ± 0.167	0.607 ± 0.021	0.105	–	0.255	–
Melanoma	0.121 ± 0.180	0.257 ± 0.006	0.309 ± 0.128	$\mathbf{0.072 \pm 0.017}$	0.079 ± 0.011	0.156 ± 0.117	0.317 ± 0.285	0.239 ± 0.080	0.098 ± 0.016	0.236 ± 0.016	0.089	–	0.095	–
PBC	0.098 ± 0.063	0.084 ± 0.025	0.193 ± 0.048	0.042 ± 0.045	$\mathbf{0.025 \pm 0.014}$	0.119 ± 0.058	0.189 ± 0.073	0.147 ± 0.295	0.103 ± 0.044	0.174 ± 0.008	0.052	–	0.067	–
PBC3	0.110 ± 0.124	0.216 ± 0.066	0.243 ± 0.048	$\mathbf{0.073 \pm 0.015}$	0.205 ± 0.053	0.178 ± 0.041	0.347 ± 0.210	0.302 ± 0.049	0.103 ± 0.027	0.360 ± 0.006	0.112	–	0.134	0.162
Retinopathy	0.064 ± 0.062	0.101 ± 0.087	0.370 ± 0.069	0.058 ± 0.027	$\mathbf{0.033 \pm 0.016}$	0.107 ± 0.050	0.164 ± 0.162	0.138 ± 0.025	0.105 ± 0.053	0.174 ± 0.014	0.043	–	0.085	–
WHAS500	0.075 ± 0.098	0.182 ± 0.072	0.244 ± 0.024	$\mathbf{0.040 \pm 0.005}$	0.091 ± 0.034	0.092 ± 0.041	0.230 ± 0.154	0.185 ± 0.060	0.065 ± 0.016	0.268 ± 0.017	–	–	0.094	0.076
GBSG2	$\mathbf{0.037 \pm 0.028}$	0.290 ± 0.109	0.400 ± 0.060	0.132 ± 0.018	0.083 ± 0.029	0.286 ± 0.231	0.397 ± 0.292	0.321 ± 0.052	0.076 ± 0.012	0.570 ± 0.040	0.214	–	0.145	–
Colon	0.053 ± 0.036	0.100 ± 0.021	0.226 ± 0.063	0.035 ± 0.014	$\mathbf{0.023 \pm 0.005}$	0.161 ± 0.150	0.254 ± 0.205	0.102 ± 0.028	0.067 ± 0.027	0.320 ± 0.027	0.089	–	0.080	–
GRACE	0.068 ± 0.037	0.190 ± 0.049	0.302 ± 0.057	$\mathbf{0.040 \pm 0.013}$	0.044 ± 0.015	0.185 ± 0.116	0.363 ± 0.481	0.160 ± 0.028	0.091 ± 0.063	0.241 ± 0.024	–	–	0.066	–
TRACE	0.039 ± 0.055	0.142 ± 0.066	0.227 ± 0.111	0.024 ± 0.012	$\mathbf{0.017 \pm 0.003}$	0.110 ± 0.041	0.230 ± 0.058	0.190 ± 0.231	0.086 ± 0.036	0.285 ± 0.013	–	–	0.059	–
Framingham	0.053 ± 0.027	0.174 ± 0.090	0.171 ± 0.035	0.035 ± 0.013	$\mathbf{0.017 \pm 0.004}$	0.132 ± 0.066	0.436 ± 0.021	0.297 ± 0.357	0.122 ± 0.056	0.316 ± 0.018	–	–	0.073	–
Blood Transfusion	0.116 ± 0.169	0.327 ± 0.185	0.396 ± 0.072	0.131 ± 0.038	$\mathbf{0.085 \pm 0.021}$	0.336 ± 0.008	0.458 ± 0.259	0.280 ± 0.087	0.133 ± 0.064	0.537 ± 0.029	–	–	0.273	–
QSAR Biodeg	$\mathbf{0.097 \pm 0.041}$	0.209 ± 0.022	0.204 ± 0.021	0.190 ± 0.031	0.425 ± 0.035	0.131 ± 0.030	0.412 ± 0.171	0.206 ± 0.019	0.105 ± 0.005	0.573 ± 0.015	–	–	0.105	–
Maternal Risk	$\mathbf{0.056 \pm 0.017}$	0.368 ± 0.017	0.352 ± 0.033	0.268 ± 0.040	0.291 ± 0.089	0.291 ± 0.258	0.435 ± 0.263	0.313 ± 0.029	0.144 ± 0.039	0.454 ± 0.010	0.061	–	0.273	0.211
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 6: KL divergence (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.544 ± 0.945	0.417 ± 0.141	5.073 ± 2.677	0.218 ± 0.021	0.889	0.739 ± 0.149	0.988 ± 1.095	6.855 ± 1.418	0.261 ± 0.126	0.548 ± 0.087	0.413	**0.152**	0.720	0.573
Breast Cancer	$\mathbf{0.108 \pm 0.064}$	0.531 ± 0.050	1.804 ± 0.285	0.260 ± 0.236	1.136	0.522 ± 0.361	6.607 ± 11.311	11.996 ± 1.070	0.173 ± 0.047	0.801 ± 0.102	–	–	2.490	–
Liver Disorders	0.483 ± 0.275	0.889 ± 0.385	3.393 ± 2.232	0.304 ± 0.132	**0.261**	1.205 ± 0.209	0.703 ± 1.668	11.831 ± 0.843	0.367 ± 0.232	0.887 ± 0.088	0.795	–	1.033	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Parkinsons	**0.409 ± 0.106**	1.241 ± 0.611	2.603 ± 0.351	0.430 ± 0.264	1.399	1.338 ± 0.547	0.973 ± 1.641	15.105 ± 2.132	0.508 ± 0.330	0.766 ± 0.056	–	–	3.060	–
Obesity	**0.033 ± 0.026**	0.122 ± 0.045	1.345 ± 0.151	0.233 ± 0.141	0.124	0.360 ± 0.562	0.456 ± 0.765	4.246 ± 0.378	0.101 ± 0.038	0.568 ± 0.006	–	–	0.375	–
Diabetes	0.194 ± 0.201	0.079 ± 0.006	0.360 ± 0.468	0.175 ± 0.041	**0.006 ± 0.001**	0.380 ± 0.574	0.584 ± 0.449	2.691 ± 0.966	0.093 ± 0.045	0.570 ± 0.056	0.235	–	0.119	0.953
German Credit	2.121 ± 2.165	0.058 ± 0.035	3.982 ± 0.208	0.036 ± 0.017	**0.024 ± 0.005**	0.142 ± 0.176	0.180 ± 0.244	1.708 ± 0.256	0.076 ± 0.052	0.368 ± 0.027	–	–	0.093	–
Bank Marketing	0.101 ± 0.062	0.028 ± 0.012	0.615 ± 0.457	0.111 ± 0.106	**0.014 ± 0.003**	0.235 ± 0.192	0.300 ± 0.530	2.197 ± 1.182	0.133 ± 0.087	0.819 ± 0.139	0.243	–	0.094	–
Census Income	0.676 ± 0.496	0.062 ± 0.018	0.987 ± 0.673	0.155 ± 0.086	**0.031 ± 0.014**	0.463 ± 0.390	1.749 ± 6.259	3.233 ± 3.534	0.257 ± 0.069	1.137 ± 0.033	0.294	–	0.105	–
Melanoma	0.630 ± 1.660	0.665 ± 0.551	3.288 ± 3.752	0.245 ± 0.161	0.227 ± 0.221	0.812 ± 1.141	1.628 ± 4.267	8.892 ± 1.341	**0.200 ± 0.149**	0.343 ± 0.092	0.383	–	0.327	–
PBC	0.384 ± 0.831	0.059 ± 0.049	2.044 ± 0.410	0.025 ± 0.017	**0.018 ± 0.008**	0.183 ± 0.314	0.157 ± 0.086	1.650 ± 3.313	0.087 ± 0.091	0.175 ± 0.028	0.063	–	0.408	–
PBC3	0.329 ± 0.517	0.323 ± 0.159	3.822 ± 1.163	**0.180 ± 0.084**	0.727 ± 0.314	0.488 ± 0.077	1.163 ± 2.106	7.724 ± 1.264	0.231 ± 0.188	0.502 ± 0.071	0.511	–	0.748	1.113
Retinopathy	2.329 ± 9.415	0.054 ± 0.073	4.137 ± 1.474	0.026 ± 0.011	**0.017 ± 0.015**	0.106 ± 0.085	0.178 ± 0.227	2.314 ± 0.104	0.059 ± 0.027	0.092 ± 0.022	0.018	–	0.104	–
WHAS500	**0.063 ± 0.078**	0.291 ± 0.234	3.193 ± 1.167	0.091 ± 0.039	0.233 ± 0.085	0.241 ± 0.108	0.495 ± 0.674	5.838 ± 0.587	0.106 ± 0.104	0.339 ± 0.018	–	–	1.104	0.291
GBSG2	0.168 ± 0.193	0.311 ± 0.276	4.253 ± 1.033	**0.093 ± 0.049**	0.115 ± 0.053	0.586 ± 0.792	0.609 ± 1.635	4.183 ± 1.379	0.138 ± 0.071	0.780 ± 0.034	1.170	–	0.215	–
Colon	0.121 ± 0.071	0.104 ± 0.067	2.855 ± 0.867	0.027 ± 0.006	**0.016 ± 0.003**	0.273 ± 0.301	0.616 ± 0.901	2.297 ± 1.226	0.031 ± 0.010	0.350 ± 0.047	1.109	–	0.057	–
GRACE	0.109 ± 0.126	0.338 ± 0.247	1.602 ± 1.417	**0.041 ± 0.015**	0.088 ± 0.186	0.862 ± 1.509	2.834 ± 10.957	6.324 ± 1.459	0.085 ± 0.060	0.384 ± 0.060	–	–	0.224	–
TRACE	0.069 ± 0.051	0.154 ± 0.095	2.590 ± 4.084	0.024 ± 0.003	**0.009 ± 0.010**	0.304 ± 0.308	0.298 ± 0.127	3.033 ± 1.847	0.059 ± 0.026	0.331 ± 0.009	–	–	0.078	–
Framingham	0.360 ± 0.077	0.157 ± 0.139	1.271 ± 0.872	0.044 ± 0.023	**0.011 ± 0.003**	0.407 ± 0.393	1.058 ± 0.287	3.481 ± 6.399	0.114 ± 0.039	0.525 ± 0.027	–	–	0.113	–
Blood Transfusion	1.006 ± 2.150	0.563 ± 0.252	3.656 ± 1.451	**0.162 ± 0.038**	0.199 ± 0.234	2.008 ± 0.824	1.265 ± 1.770	11.252 ± 1.432	0.251 ± 0.073	0.906 ± 0.074	–	–	0.421	–
QSAR Biodeg	0.567 ± 0.189	0.263 ± 0.076	2.156 ± 0.388	1.629 ± 1.281	10.681 ± 1.055	0.296 ± 0.175	3.983 ± 4.591	3.813 ± 0.579	**0.180 ± 0.084**	1.125 ± 0.028	–	–	0.465	–
Maternal Risk	0.621 ± 0.382	1.003 ± 0.195	5.678 ± 1.994	1.110 ± 0.781	1.200 ± 0.729	1.545 ± 1.718	3.575 ± 4.765	12.353 ± 0.949	0.412 ± 0.222	1.023 ± 0.059	**0.299**	–	1.232	0.804
Website Phishing	0.854 ± 3.562	5.346 ± 0.005	5.952 ± 0.319	5.343 ± 0.002	**0.002 ± 0.002**	0.013 ± 0.017	0.194 ± 0.199	0.009 ± 0.014	0.022 ± 0.011	0.074 ± 0.012	–	–	0.008	–

Table 7: Hellinger (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.161 ± 0.075	0.208 ± 0.010	0.419 ± 0.047	0.137 ± 0.008	0.326	0.195 ± 0.025	0.319 ± 0.127	0.376 ± 0.039	0.147 ± 0.013	0.322 ± 0.018	0.147	**0.087**	0.189	0.156
Breast Cancer	**0.097 ± 0.054**	0.313 ± 0.013	0.297 ± 0.028	0.138 ± 0.015	0.490	0.219 ± 0.031	0.668 ± 0.263	0.636 ± 0.019	0.132 ± 0.033	0.477 ± 0.027	–	–	0.398	–
Liver Disorders	0.153 ± 0.021	0.360 ± 0.045	0.331 ± 0.090	0.147 ± 0.013	**0.147**	0.290 ± 0.047	0.281 ± 0.104	0.633 ± 0.042	0.176 ± 0.011	0.501 ± 0.018	0.180	–	0.300	–
Parkinsons	**0.149 ± 0.033**	0.357 ± 0.033	0.333 ± 0.018	0.208 ± 0.028	0.427	0.301 ± 0.022	0.320 ± 0.186	0.718 ± 0.051	0.195 ± 0.010	0.453 ± 0.017	–	–	0.421	–
Obesity	**0.068 ± 0.023**	0.149 ± 0.022	0.245 ± 0.024	0.133 ± 0.011	0.154	0.143 ± 0.126	0.241 ± 0.170	0.277 ± 0.014	0.128 ± 0.028	0.367 ± 0.007	–	–	0.152	–
Diabetes	0.106 ± 0.032	0.135 ± 0.011	0.142 ± 0.028	0.148 ± 0.005	**0.027 ± 0.007**	0.161 ± 0.056	0.302 ± 0.127	0.208 ± 0.060	0.133 ± 0.038	0.402 ± 0.020	0.160	–	0.116	0.238
German Credit	0.303 ± 0.577	0.087 ± 0.010	0.368 ± 0.015	**0.043 ± 0.005**	0.049 ± 0.002	0.082 ± 0.009	0.126 ± 0.043	0.138 ± 0.011	0.112 ± 0.016	0.300 ± 0.010	–	–	0.080	–
Bank Marketing	0.083 ± 0.015	0.069 ± 0.021	0.163 ± 0.041	0.065 ± 0.010	**0.046 ± 0.015**	0.112 ± 0.029	0.200 ± 0.145	0.178 ± 0.048	0.140 ± 0.036	0.416 ± 0.020	0.155	–	0.074	–
Census Income	0.172 ± 0.066	0.109 ± 0.020	0.186 ± 0.037	0.111 ± 0.069	**0.062 ± 0.010**	0.194 ± 0.067	0.308 ± 0.328	0.274 ± 0.092	0.213 ± 0.050	0.491 ± 0.007	0.198	–	0.083	–
Melanoma	0.166 ± 0.170	0.249 ± 0.022	0.337 ± 0.113	0.137 ± 0.028	0.154 ± 0.042	0.200 ± 0.101	0.328 ± 0.231	0.436 ± 0.036	0.144 ± 0.019	0.269 ± 0.044	**0.120**	–	0.146	–
PBC	0.142 ± 0.139	0.081 ± 0.010	0.229 ± 0.021	0.045 ± 0.043	**0.039 ± 0.006**	0.098 ± 0.019	0.157 ± 0.078	0.153 ± 0.175	0.090 ± 0.024	0.184 ± 0.017	0.066	–	0.076	–
PBC3	0.152 ± 0.098	0.185 ± 0.018	0.326 ± 0.052	**0.106 ± 0.013**	0.294 ± 0.069	0.170 ± 0.025	0.311 ± 0.156	0.417 ± 0.041	0.142 ± 0.005	0.332 ± 0.016	0.158	–	0.176	0.215
Retinopathy	0.202 ± 0.304	0.074 ± 0.022	0.308 ± 0.049	0.050 ± 0.004	**0.040 ± 0.017**	0.105 ± 0.051	0.140 ± 0.067	0.144 ± 0.010	0.094 ± 0.005	0.115 ± 0.013	0.052	–	0.091	–
WHAS500	0.084 ± 0.068	0.172 ± 0.063	0.318 ± 0.037	**0.067 ± 0.005**	0.164 ± 0.039	0.113 ± 0.023	0.233 ± 0.122	0.332 ± 0.045	0.086 ± 0.015	0.273 ± 0.006	–	–	0.161	0.105
GBSG2	0.118 ± 0.118	0.167 ± 0.048	0.379 ± 0.021	**0.088 ± 0.009**	0.114 ± 0.021	0.206 ± 0.180	0.254 ± 0.261	0.294 ± 0.049	0.100 ± 0.024	0.396 ± 0.015	0.150	–	0.117	–
Colon	0.076 ± 0.049	0.077 ± 0.014	0.274 ± 0.071	**0.038 ± 0.018**	0.038 ± 0.008	0.145 ± 0.107	0.187 ± 0.107	0.162 ± 0.024	0.062 ± 0.017	0.262 ± 0.017	0.159	–	0.068	–
GRACE	0.085 ± 0.041	0.185 ± 0.053	0.282 ± 0.070	**0.062 ± 0.007**	0.085 ± 0.021	0.193 ± 0.069	0.341 ± 0.442	0.383 ± 0.049	0.094 ± 0.032	0.271 ± 0.008	–	–	0.122	–
TRACE	0.061 ± 0.064	0.145 ± 0.046	0.277 ± 0.114	0.048 ± 0.011	**0.032 ± 0.011**	0.148 ± 0.045	0.218 ± 0.079	0.275 ± 0.061	0.100 ± 0.021	0.282 ± 0.004	–	–	0.072	–
Framingham	0.093 ± 0.018	0.191 ± 0.094	0.206 ± 0.026	0.074 ± 0.014	**0.049 ± 0.012**	0.190 ± 0.036	0.432 ± 0.041	0.382 ± 0.152	0.150 ± 0.025	0.359 ± 0.003	–	–	0.135	–
Blood Transfusion	0.214 ± 0.219	0.303 ± 0.106	0.399 ± 0.079	0.166 ± 0.041	**0.149 ± 0.017**	0.364 ± 0.044	0.399 ± 0.075	0.609 ± 0.078	0.178 ± 0.040	0.499 ± 0.029	–	–	0.249	–
QSAR Biodeg	0.144 ± 0.029	0.190 ± 0.009	0.257 ± 0.018	0.200 ± 0.021	0.602 ± 0.008	0.144 ± 0.027	0.401 ± 0.195	0.290 ± 0.020	**0.133 ± 0.013**	0.550 ± 0.008	–	–	0.163	–
Maternal Risk	0.174 ± 0.037	0.471 ± 0.056	0.470 ± 0.022	0.427 ± 0.009	0.450 ± 0.103	0.397 ± 0.119	0.513 ± 0.162	0.618 ± 0.026	0.304 ± 0.025	0.538 ± 0.010	**0.135**	–	0.362	0.293
Website Phishing	0.167 ± 0.437	0.247 ± 0.012	0.358 ± 0.005	0.236 ± 0.007	**0.019 ± 0.009**	0.050 ± 0.023	0.184 ± 0.070	0.039 ± 0.033	0.066 ± 0.009	0.121 ± 0.010	–	–	0.041	–

Table 8: Pearson corr. preservation (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.914 ± 0.074	0.904 ± 0.030	0.926 ± 0.013	0.930 ± 0.012	0.822	0.912 ± 0.018	0.844 ± 0.042	0.918 ± 0.012	0.908 ± 0.011	0.900 ± 0.005	0.897	**0.939**	0.883	0.898
Breast Cancer	0.923 ± 0.030	0.606 ± 0.006	0.799 ± 0.009	**0.948 ± 0.018**	0.773	0.887 ± 0.039	0.566 ± 0.047	0.753 ± 0.028	0.694 ± 0.048	0.607 ± 0.012	–	–	0.746	–
Liver Disorders	0.906 ± 0.091	0.744 ± 0.034	0.831 ± 0.125	**0.948 ± 0.005**	0.912	0.862 ± 0.092	0.640 ± 0.144	0.734 ± 0.068	0.814 ± 0.021	0.743 ± 0.034	0.893	–	0.897	–
Parkinsons	0.889 ± 0.044	0.487 ± 0.042	0.653 ± 0.012	**0.936 ± 0.010**	0.739	0.814 ± 0.054	0.481 ± 0.057	0.606 ± 0.043	0.610 ± 0.052	0.486 ± 0.030	–	–	0.852	–
Obesity	**0.958 ± 0.004**	0.871 ± 0.007	0.895 ± 0.014	0.941 ± 0.014	0.847	0.953 ± 0.088	0.856 ± 0.053	0.923 ± 0.004	0.892 ± 0.022	0.874 ± 0.010	–	–	0.940	–
Diabetes	0.933 ± 0.070	0.898 ± 0.033	0.934 ± 0.037	0.916 ± 0.011	**0.986 ± 0.004**	0.968 ± 0.008	0.882 ± 0.039	0.905 ± 0.015	0.918 ± 0.032	0.868 ± 0.013	0.934	–	0.952	0.809
German Credit	0.926 ± 0.066	0.896 ± 0.015	0.885 ± 0.052	**0.933 ± 0.020**	0.923 ± 0.057	0.933 ± 0.023	0.879 ± 0.022	0.927 ± 0.029	0.905 ± 0.004	0.897 ± 0.008	–	–	0.922	–
Bank Marketing	0.964 ± 0.017	0.943 ± 0.028	0.896 ± 0.062	0.966 ± 0.022	0.824 ± 0.174	**0.969 ± 0.016**	0.922 ± 0.008	0.969 ± 0.031	0.947 ± 0.006	0.940 ± 0.002	0.959	–	0.959	–
Census Income	0.969 ± 0.021	0.953 ± 0.018	0.942 ± 0.004	**0.980 ± 0.005**	0.864 ± 0.038	0.970 ± 0.013	0.914 ± 0.030	0.961 ± 0.014	0.946 ± 0.021	0.941 ± 0.005	0.941	–	0.951	–
Melanoma	0.897 ± 0.035	0.832 ± 0.032	0.888 ± 0.035	0.872 ± 0.060	0.889 ± 0.034	0.855 ± 0.119	0.794 ± 0.028	0.848 ± 0.022	0.865 ± 0.109	0.827 ± 0.065	**0.915**	–	0.751	–
PBC	0.889 ± 0.132	0.859 ± 0.043	0.709 ± 0.220	0.897 ± 0.048	**0.941 ± 0.015**	0.924 ± 0.023	0.862 ± 0.055	0.887 ± 0.034	0.897 ± 0.063	0.861 ± 0.029	0.915	–	0.863	–
PBC3	0.912 ± 0.048	0.864 ± 0.011	0.883 ± 0.051	**0.921 ± 0.017**	0.860 ± 0.027	0.916 ± 0.030	0.857 ± 0.023	0.889 ± 0.007	0.882 ± 0.014	0.863 ± 0.021	0.857	–	0.870	0.893
Retinopathy	0.956 ± 0.051	0.781 ± 0.063	0.820 ± 0.177	0.824 ± 0.110	0.959 ± 0.060	0.868 ± 0.017	0.784 ± 0.037	0.852 ± 0.077	0.813 ± 0.027	0.779 ± 0.034	**0.983**	–	0.876	–
WHAS500	0.924 ± 0.027	0.891 ± 0.015	0.903 ± 0.044	0.931 ± 0.010	0.891 ± 0.039	**0.931 ± 0.011**	0.870 ± 0.023	0.917 ± 0.003	0.914 ± 0.010	0.897 ± 0.009	–	–	0.892	0.896
GBSG2	0.907 ± 0.126	0.853 ± 0.040	0.871 ± 0.062	0.920 ± 0.042	0.820 ± 0.186	0.882 ± 0.116	0.803 ± 0.059	0.884 ± 0.081	0.887 ± 0.048	0.848 ± 0.018	0.886	–	**0.950**	–
Colon	0.960 ± 0.023	0.921 ± 0.013	0.807 ± 0.061	0.942 ± 0.032	**0.961 ± 0.015**	0.915 ± 0.062	0.911 ± 0.004	0.947 ± 0.005	0.931 ± 0.009	0.917 ± 0.013	0.946	–	0.957	–
GRACE	**0.961 ± 0.024**	0.916 ± 0.036	0.873 ± 0.061	0.958 ± 0.006	0.947 ± 0.061	0.942 ± 0.031	0.883 ± 0.074	0.918 ± 0.014	0.950 ± 0.036	0.919 ± 0.041	–	–	0.893	–
TRACE	0.970 ± 0.016	0.870 ± 0.003	0.907 ± 0.023	0.930 ± 0.005	**0.982 ± 0.007**	0.931 ± 0.058	0.853 ± 0.021	0.877 ± 0.046	0.924 ± 0.057	0.876 ± 0.009	–	–	0.930	–
Framingham	0.972 ± 0.005	0.846 ± 0.031	0.740 ± 0.050	**0.983 ± 0.004**	0.968 ± 0.026	0.964 ± 0.004	0.792 ± 0.016	0.887 ± 0.017	0.883 ± 0.025	0.833 ± 0.006	–	–	0.902	–
Blood Transfusion	0.748 ± 0.482	0.548 ± 0.016	0.729 ± 0.060	0.931 ± 0.038	**0.969 ± 0.051**	0.880 ± 0.110	0.576 ± 0.145	0.680 ± 0.163	0.731 ± 0.142	0.549 ± 0.092	–	–	0.882	–
QSAR Biodeg	0.821 ± 0.025	0.829 ± 0.002	0.856 ± 0.036	**0.912 ± 0.013**	0.822 ± 0.026	0.906 ± 0.010	0.809 ± 0.025	0.873 ± 0.002	0.837 ± 0.014	0.829 ± 0.003	–	–	0.868	–
Maternal Risk	0.922 ± 0.029	0.755 ± 0.008	0.832 ± 0.035	**0.929 ± 0.017**	0.794 ± 0.155	0.906 ± 0.081	0.751 ± 0.036	0.882 ± 0.039	0.822 ± 0.033	0.750 ± 0.035	0.832	–	0.883	0.701
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 9: JS numeric (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.042 ± 0.035	0.094 ± 0.020	0.200 ± 0.033	0.042 ± 0.002	0.194	0.070 ± 0.006	0.192 ± 0.132	0.265 ± 0.048	0.040 ± 0.004	0.189 ± 0.014	0.049	**0.013**	0.069	0.046
Breast Cancer	**0.013 ± 0.014**	0.143 ± 0.007	0.111 ± 0.022	0.027 ± 0.011	0.294	0.072 ± 0.019	0.568 ± 0.357	0.466 ± 0.024	0.024 ± 0.014	0.293 ± 0.032	–	–	0.210	–
Liver Disorders	0.031 ± 0.009	0.184 ± 0.038	0.129 ± 0.058	0.028 ± 0.005	**0.028**	0.113 ± 0.031	0.118 ± 0.095	0.450 ± 0.074	0.040 ± 0.004	0.316 ± 0.019	0.042	–	0.118	–
Parkinsons	**0.028 ± 0.014**	0.175 ± 0.029	0.136 ± 0.013	0.056 ± 0.013	0.227	0.120 ± 0.015	0.143 ± 0.175	0.578 ± 0.075	0.049 ± 0.004	0.269 ± 0.016	–	–	0.221	–
Obesity	**0.013 ± 0.006**	0.081 ± 0.036	0.136 ± 0.030	0.114 ± 0.022	0.072	0.093 ± 0.144	0.237 ± 0.334	0.313 ± 0.013	0.052 ± 0.014	0.206 ± 0.008	–	–	0.101	–
Diabetes	0.021 ± 0.009	0.033 ± 0.003	0.030 ± 0.011	0.089 ± 0.009	**0.002 ± 0.001**	0.082 ± 0.054	0.216 ± 0.114	0.125 ± 0.072	0.036 ± 0.022	0.248 ± 0.024	0.034	–	0.040	0.124
German Credit	0.036 ± 0.029	0.047 ± 0.021	0.202 ± 0.060	0.023 ± 0.009	**0.017 ± 0.001**	0.044 ± 0.004	0.111 ± 0.054	0.181 ± 0.026	0.031 ± 0.014	0.160 ± 0.013	–	–	0.043	–
Bank Marketing	0.011 ± 0.007	0.010 ± 0.006	0.043 ± 0.030	0.035 ± 0.013	**0.007 ± 0.007**	0.027 ± 0.013	0.098 ± 0.163	0.123 ± 0.003	0.025 ± 0.015	0.454 ± 0.076	0.045	–	0.031	–
Census Income	0.052 ± 0.076	0.020 ± 0.008	0.071 ± 0.036	0.115 ± 0.139	**0.010 ± 0.003**	0.069 ± 0.034	0.188 ± 0.498	0.139 ± 0.098	0.046 ± 0.023	0.445 ± 0.025	0.029	–	0.021	–
Melanoma	0.046 ± 0.083	0.118 ± 0.005	0.147 ± 0.076	0.038 ± 0.013	0.044 ± 0.022	0.077 ± 0.079	0.187 ± 0.239	0.331 ± 0.072	0.033 ± 0.005	0.127 ± 0.033	**0.020**	–	0.040	–
PBC	0.024 ± 0.043	0.026 ± 0.022	0.074 ± 0.014	**0.006 ± 0.008**	0.007 ± 0.003	0.025 ± 0.014	0.062 ± 0.012	0.111 ± 0.198	0.016 ± 0.012	0.058 ± 0.005	0.007	–	0.023	–
PBC3	0.035 ± 0.050	0.076 ± 0.006	0.126 ± 0.042	**0.024 ± 0.011**	0.166 ± 0.064	0.055 ± 0.010	0.197 ± 0.181	0.330 ± 0.047	0.030 ± 0.003	0.194 ± 0.016	0.043	–	0.061	0.079
Retinopathy	0.013 ± 0.012	0.032 ± 0.042	0.269 ± 0.021	0.016 ± 0.007	0.009 ± 0.007	0.036 ± 0.023	0.065 ± 0.082	0.159 ± 0.013	0.031 ± 0.020	0.054 ± 0.012	**0.009**	–	0.032	–
WHAS500	0.013 ± 0.022	0.075 ± 0.051	0.131 ± 0.032	**0.011 ± 0.001**	0.063 ± 0.029	0.030 ± 0.013	0.129 ± 0.116	0.244 ± 0.020	0.015 ± 0.005	0.129 ± 0.010	–	–	0.066	0.022
GBSG2	**0.012 ± 0.009**	0.098 ± 0.071	0.193 ± 0.039	0.023 ± 0.010	0.035 ± 0.012	0.109 ± 0.127	0.198 ± 0.306	0.260 ± 0.092	0.022 ± 0.004	0.392 ± 0.027	0.069	–	0.039	–
Colon	0.010 ± 0.010	0.026 ± 0.016	0.103 ± 0.054	**0.004 ± 0.001**	0.006 ± 0.002	0.056 ± 0.077	0.128 ± 0.102	0.110 ± 0.024	0.009 ± 0.006	0.145 ± 0.019	0.015	–	0.014	–
GRACE	0.011 ± 0.007	0.085 ± 0.052	0.120 ± 0.043	**0.008 ± 0.001**	0.013 ± 0.005	0.069 ± 0.042	0.225 ± 0.495	0.269 ± 0.055	0.015 ± 0.012	0.149 ± 0.019	–	–	0.031	–
TRACE	0.009 ± 0.012	0.055 ± 0.032	0.111 ± 0.078	0.009 ± 0.001	**0.003 ± 0.002**	0.058 ± 0.041	0.100 ± 0.047	0.180 ± 0.130	0.020 ± 0.009	0.134 ± 0.003	–	–	0.013	–
Framingham	0.017 ± 0.002	0.059 ± 0.052	0.095 ± 0.012	0.010 ± 0.003	**0.004 ± 0.002**	0.060 ± 0.020	0.297 ± 0.057	0.226 ± 0.165	0.035 ± 0.011	0.205 ± 0.008	–	–	0.029	–
Blood Transfusion	0.067 ± 0.130	0.135 ± 0.061	0.193 ± 0.072	0.037 ± 0.018	**0.029 ± 0.004**	0.173 ± 0.049	0.228 ± 0.092	0.425 ± 0.105	0.043 ± 0.020	0.333 ± 0.025	–	–	0.085	–
QSAR Biodeg	0.033 ± 0.003	0.065 ± 0.014	0.090 ± 0.010	0.094 ± 0.027	0.509 ± 0.001	0.039 ± 0.016	0.272 ± 0.226	0.144 ± 0.022	**0.029 ± 0.005**	0.391 ± 0.011	–	–	0.049	–
Maternal Risk	0.038 ± 0.017	0.289 ± 0.069	0.264 ± 0.025	0.249 ± 0.020	0.275 ± 0.119	0.235 ± 0.132	0.362 ± 0.175	0.476 ± 0.047	0.140 ± 0.020	0.360 ± 0.014	**0.025**	–	0.167	0.114
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 10: JS categorical (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Breast Cancer	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Liver Disorders	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Parkinsons	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Obesity	0.004 ± 0.003	0.006 ± 0.003	0.044 ± 0.001	**0.000 ± 0.000**	0.015	0.002 ± 0.005	0.015 ± 0.018	0.002 ± 0.003	0.014 ± 0.018	0.195 ± 0.012	–	–	0.002	–
Diabetes	0.010 ± 0.015	0.020 ± 0.013	0.049 ± 0.025	**0.000 ± 0.000**	0.000 ± 0.000	0.003 ± 0.005	0.047 ± 0.104	0.002 ± 0.005	0.022 ± 0.028	0.155 ± 0.009	0.144	–	0.003	0.017
German Credit	0.290 ± 0.844	0.005 ± 0.001	0.153 ± 0.023	**0.001 ± 0.000**	0.001 ± 0.000	0.005 ± 0.000	0.011 ± 0.010	0.006 ± 0.002	0.019 ± 0.005	0.129 ± 0.007	–	–	0.004	–
Bank Marketing	0.014 ± 0.005	0.008 ± 0.004	0.038 ± 0.012	**0.000 ± 0.000**	0.003 ± 0.001	0.050 ± 0.027	0.079 ± 0.070	0.052 ± 0.061	0.051 ± 0.040	0.135 ± 0.003	0.055	–	0.003	–
Census Income	0.049 ± 0.032	0.018 ± 0.014	0.043 ± 0.014	**0.000 ± 0.000**	0.006 ± 0.003	0.092 ± 0.109	0.177 ± 0.239	0.150 ± 0.195	0.114 ± 0.036	0.274 ± 0.003	0.095	–	0.005	–
Melanoma	–	–	–	–	–	–	–	–	–	–	–	–	–	–
PBC	0.129 ± 0.460	0.004 ± 0.004	0.058 ± 0.008	0.002 ± 0.004	**0.000 ± 0.001**	0.004 ± 0.006	0.012 ± 0.019	0.007 ± 0.016	0.005 ± 0.011	0.129 ± 0.048	0.022	–	0.005	–
PBC3	0.077 ± 0.039	0.003 ± 0.005	0.307 ± 0.035	**0.001 ± 0.002**	0.028 ± 0.051	0.008 ± 0.016	0.023 ± 0.037	0.008 ± 0.013	0.050 ± 0.013	0.007 ± 0.007	0.014	–	0.003	0.044
Retinopathy	0.270 ± 0.868	0.001 ± 0.001	0.034 ± 0.051	**0.000 ± 0.000**	0.001 ± 0.001	0.011 ± 0.025	0.013 ± 0.017	0.001 ± 0.005	0.006 ± 0.006	0.003 ± 0.003	0.001	–	0.000	–
WHAS500	–	–	–	–	–	–	–	–	–	–	–	–	–	–
GBSG2	0.061 ± 0.202	0.002 ± 0.004	0.139 ± 0.009	**0.001 ± 0.001**	0.001 ± 0.001	0.033 ± 0.083	0.018 ± 0.070	0.001 ± 0.001	0.005 ± 0.013	0.035 ± 0.016	0.007	–	0.001	–
Colon	0.012 ± 0.022	0.002 ± 0.001	0.108 ± 0.045	0.001 ± 0.001	**0.001 ± 0.001**	0.032 ± 0.034	0.018 ± 0.019	0.006 ± 0.006	0.007 ± 0.010	0.087 ± 0.010	0.241	–	0.005	–
GRACE	–	–	–	–	–	–	–	–	–	–	–	–	–	–
TRACE	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Framingham	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Blood Transfusion	–	–	–	–	–	–	–	–	–	–	–	–	–	–
QSAR Biodeg	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Maternal Risk	–	–	–	–	–	–	–	–	–	–	–	–	–	–
Website Phishing	0.100 ± 0.394	0.224 ± 0.001	0.259 ± 0.003	0.223 ± 0.000	**0.001 ± 0.001**	0.005 ± 0.006	0.060 ± 0.052	0.003 ± 0.005	0.008 ± 0.005	0.028 ± 0.004	–	–	0.003	–

Privacy / disclosure

Table 11: DCR median (↑=less memorization) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.715 ± 0.026	0.869 ± 0.158	0.553 ± 0.234	0.736 ± 0.007	**1.971**	0.703 ± 0.090	1.043 ± 0.312	0.617 ± 0.114	0.715 ± 0.050	1.873 ± 0.111	0.655	0.684	0.611	0.623
Breast Cancer	0.559 ± 0.019	1.245 ± 0.120	0.476 ± 0.047	0.555 ± 0.037	**2.769**	0.672 ± 0.083	1.726 ± 0.372	0.433 ± 0.045	0.850 ± 0.176	2.494 ± 0.208	–	–	0.508	–
Liver Disorders	0.359 ± 0.104	0.682 ± 0.169	0.240 ± 0.013	0.382 ± 0.048	0.307	0.316 ± 0.036	0.570 ± 0.144	0.379 ± 0.087	0.376 ± 0.124	**1.677 ± 0.319**	0.273	–	0.240	–
Parkinsons	0.511 ± 0.027	1.033 ± 0.097	0.510 ± 0.052	0.538 ± 0.063	0.928	0.625 ± 0.124	1.019 ± 0.193	0.457 ± 0.028	0.742 ± 0.102	**1.923 ± 0.271**	–	–	0.523	–
Obesity	0.281 ± 0.026	0.437 ± 0.042	0.247 ± 0.025	0.403 ± 0.032	0.325	0.342 ± 0.088	0.471 ± 0.041	0.349 ± 0.023	0.426 ± 0.019	**0.706 ± 0.031**	–	–	0.299	–
Diabetes	0.053 ± 0.024	0.082 ± 0.021	0.038 ± 0.009	0.116 ± 0.021	0.056 ± 0.001	0.065 ± 0.008	0.118 ± 0.169	0.068 ± 0.041	0.081 ± 0.021	**0.545 ± 0.125**	0.033	–	0.047	0.109
German Credit	0.432 ± 0.091	0.456 ± 0.004	0.308 ± 0.040	0.426 ± 0.008	0.392 ± 0.005	0.418 ± 0.023	0.431 ± 0.001	0.402 ± 0.022	0.446 ± 0.024	**0.642 ± 0.028**	–	–	0.387	–
Bank Marketing	0.179 ± 0.055	0.254 ± 0.020	0.096 ± 0.006	0.269 ± 0.015	0.211 ± 0.002	0.250 ± 0.039	0.331 ± 0.128	0.226 ± 0.033	0.289 ± 0.037	**2.937 ± 5.289**	0.199	–	0.164	–
Census Income	0.102 ± 0.064	0.195 ± 0.012	0.044 ± 0.007	0.254 ± 0.137	0.140 ± 0.000	0.188 ± 0.056	0.247 ± 0.050	0.189 ± 0.077	0.222 ± 0.039	**0.683 ± 0.097**	0.098	–	0.079	–
Melanoma	0.367 ± 0.050	0.655 ± 0.209	0.322 ± 0.130	0.404 ± 0.106	0.360 ± 0.028	0.375 ± 0.047	0.748 ± 0.091	0.348 ± 0.097	0.419 ± 0.084	**0.891 ± 0.076**	0.353	–	0.414	–
PBC	0.224 ± 0.620	0.201 ± 0.064	0.080 ± 0.065	0.130 ± 0.105	0.080 ± 0.006	0.075 ± 0.031	0.158 ± 0.152	0.242 ± 0.773	0.084 ± 0.007	**0.571 ± 0.057**	0.074	–	0.061	–
PBC3	0.566 ± 0.021	0.722 ± 0.037	0.457 ± 0.013	0.602 ± 0.006	1.112 ± 0.474	0.600 ± 0.044	0.809 ± 0.259	0.464 ± 0.025	0.575 ± 0.060	**1.414 ± 0.109**	0.524	–	0.514	0.508
Retinopathy	0.345 ± 0.316	0.472 ± 0.055	0.111 ± 0.107	0.372 ± 0.134	0.259 ± 0.114	0.300 ± 0.012	0.454 ± 0.134	0.321 ± 0.021	0.373 ± 0.161	**0.542 ± 0.036**	0.270	–	0.283	–
WHAS500	0.653 ± 0.083	0.934 ± 0.105	0.480 ± 0.138	0.722 ± 0.026	0.725 ± 0.036	0.678 ± 0.070	1.125 ± 0.123	0.571 ± 0.116	0.702 ± 0.018	**1.822 ± 0.304**	–	–	0.619	0.677
GBSG2	0.291 ± 0.121	0.450 ± 0.042	0.124 ± 0.012	0.325 ± 0.025	0.251 ± 0.027	0.324 ± 0.187	0.534 ± 0.309	0.261 ± 0.049	0.334 ± 0.046	**1.615 ± 0.146**	0.242	–	0.218	–
Colon	0.204 ± 0.094	0.368 ± 0.082	0.074 ± 0.006	0.258 ± 0.026	0.193 ± 0.051	0.271 ± 0.292	0.430 ± 0.059	0.142 ± 0.079	0.206 ± 0.107	**1.057 ± 0.203**	0.087	–	0.138	–
GRACE	0.221 ± 0.004	0.354 ± 0.236	0.177 ± 0.029	0.233 ± 0.028	0.221 ± 0.019	0.207 ± 0.032	0.676 ± 0.890	0.191 ± 0.019	0.209 ± 0.014	**0.975 ± 0.106**	–	–	0.206	–
TRACE	0.084 ± 0.049	0.233 ± 0.031	0.069 ± 0.073	0.101 ± 0.001	0.087 ± 0.009	0.072 ± 0.025	0.230 ± 0.356	0.238 ± 0.562	0.085 ± 0.017	**0.768 ± 0.023**	–	–	0.072	–
Framingham	0.314 ± 0.009	0.467 ± 0.003	0.276 ± 0.016	0.325 ± 0.008	0.308 ± 0.010	0.330 ± 0.051	0.724 ± 0.431	0.466 ± 0.464	0.342 ± 0.036	**1.356 ± 0.084**	–	–	0.294	–
Blood Transfusion	0.216 ± 0.109	0.542 ± 0.116	0.146 ± 0.025	0.196 ± 0.042	0.153 ± 0.017	0.172 ± 0.069	0.313 ± 0.135	0.244 ± 0.035	0.230 ± 0.058	**2.031 ± 0.130**	–	–	0.196	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
QSAR Biodeg	0.490 ± 0.030	1.089 ± 0.089	0.389 ± 0.022	0.995 ± 0.356	$\mathbf{5.396 \pm 1.573}$	0.591 ± 0.164	1.033 ± 0.515	0.482 ± 0.014	0.829 ± 0.365	4.355 ± 0.251	–	–	0.665	–
Maternal Risk	0.324 ± 0.063	0.653 ± 0.325	0.256 ± 0.083	0.442 ± 0.011	1.012 ± 0.821	0.353 ± 0.079	0.486 ± 0.099	0.307 ± 0.084	0.440 ± 0.081	$\mathbf{1.381 \pm 0.189}$	0.331	–	0.315	0.326
Website Phishing	0.322 ± 0.168	$\mathbf{0.400 \pm 0.000}$	$\mathbf{0.400 \pm 0.000}$	$\mathbf{0.400 \pm 0.000}$	0.283 ± 0.000	0.283 ± 0.000	$\mathbf{0.400 \pm 0.000}$	0.283 ± 0.000	0.322 ± 0.168	$\mathbf{0.400 \pm 0.000}$	–	–	0.283	–

Table 12: DCR real–real reference — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.796	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.801 ± 0.024	0.796	0.796	0.796	0.796
Breast Cancer	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.468	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	0.465 ± 0.057	–	–	0.468	–
Liver Disorders	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.342	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.380 ± 0.131	0.342	–	0.342	–
Parkinsons	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.538	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	0.524 ± 0.058	–	–	0.538	–
Obesity	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.284	0.276 ± 0.129	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	0.279 ± 0.028	–	–	0.284	–
Diabetes	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071 ± 0.000	0.071	–	0.071	0.071
German Credit	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	0.419 ± 0.015	–	–	0.423	–
Bank Marketing	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.209 ± 0.002	0.210	–	0.210	–
Census Income	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136 ± 0.001	0.136	–	0.136	–
Melanoma	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.524 ± 0.041	0.543	–	0.543	–
PBC	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.196 ± 0.163	0.271	–	0.271	–
PBC3	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.626 ± 0.053	0.629	–	0.629	0.629
Retinopathy	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.318 ± 0.016	0.326	–	0.326	–
WHAS500	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	0.745 ± 0.023	–	–	0.741	0.741
GBSG2	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.323 ± 0.105	0.371	–	0.371	–
Colon	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.337 ± 0.202	0.384	–	0.384	–
GRACE	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	0.290 ± 0.029	–	–	0.303	–
TRACE	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	0.109 ± 0.006	–	–	0.110	–
Framingham	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	0.352 ± 0.009	–	–	0.355	–
Blood Transfusion	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	0.201 ± 0.026	–	–	0.212	–
QSAR Biodeg	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	0.436 ± 0.030	–	–	0.440	–
Maternal Risk	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.357 ± 0.174	0.356	–	0.356	0.356
Website Phishing	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	0.400 ± 0.000	–	–	0.400	–

Table 13: DCR encoded (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	2.087 ± 0.579	2.316 ± 0.299	3.165 ± 0.177	1.999 ± 0.143	2.022	2.025 ± 0.662	2.880 ± 0.566	1.582 ± 0.178	1.918 ± 0.340	$\mathbf{3.810 \pm 0.402}$	1.769	1.669	1.879	1.860
Breast Cancer	3.263 ± 0.423	5.484 ± 0.782	3.199 ± 0.489	2.429 ± 0.233	2.308	3.052 ± 0.345	6.717 ± 3.045	2.951 ± 0.615	3.495 ± 0.493	$\mathbf{12.169 \pm 0.265}$	–	–	3.282	–
Liver Disorders	0.604 ± 0.108	0.989 ± 0.150	0.570 ± 0.047	0.560 ± 0.121	0.470	0.676 ± 0.243	0.797 ± 0.059	0.937 ± 0.602	0.594 ± 0.033	$\mathbf{1.627 \pm 0.322}$	0.502	–	0.666	–
Parkinsons	2.407 ± 0.153	3.982 ± 0.120	2.595 ± 0.205	1.978 ± 0.149	1.756	2.676 ± 0.238	3.538 ± 0.665	2.529 ± 0.171	2.767 ± 0.446	$\mathbf{8.250 \pm 1.029}$	–	–	2.645	–
Obesity	1.196 ± 0.154	1.909 ± 0.085	1.959 ± 0.111	1.860 ± 0.310	1.273	1.516 ± 0.066	2.083 ± 0.472	1.459 ± 0.129	1.770 ± 0.063	$\mathbf{3.013 \pm 0.145}$	–	–	1.451	–
Diabetes	0.480 ± 0.039	0.558 ± 0.027	0.546 ± 0.076	0.660 ± 0.105	–	0.517 ± 0.052	0.822 ± 0.185	0.524 ± 0.106	0.568 ± 0.060	$\mathbf{1.604 \pm 0.012}$	0.720	–	0.491	1.485
German Credit	3.186 ± 0.970	3.107 ± 0.154	3.516 ± 0.132	3.007 ± 0.024	–	2.914 ± 0.062	3.187 ± 0.138	2.880 ± 0.062	3.105 ± 0.140	$\mathbf{4.129 \pm 0.139}$	–	–	2.891	–
Bank Marketing	1.659 ± 0.034	1.951 ± 0.121	1.893 ± 0.243	2.006 ± 0.020	–	1.994 ± 0.168	2.382 ± 0.464	1.959 ± 0.405	2.111 ± 0.111	$\mathbf{5.609 \pm 1.075}$	1.974	–	1.651	–
Census Income	1.653 ± 0.124	2.048 ± 0.035	1.781 ± 0.231	2.226 ± 0.404	–	2.049 ± 0.320	2.686 ± 0.876	2.221 ± 0.626	2.212 ± 0.155	$\mathbf{4.370 \pm 0.304}$	1.982	–	1.580	–
Melanoma	0.570 ± 0.178	0.887 ± 0.172	$\mathbf{1.618 \pm 1.285}$	0.546 ± 0.127	0.531 ± 0.022	0.597 ± 0.283	1.072 ± 0.963	0.865 ± 0.360	0.525 ± 0.056	0.904 ± 0.178	0.508	–	0.688	–
PBC	0.605 ± 1.356	0.074 ± 0.048	$\mathbf{0.961 \pm 1.641}$	0.070 ± 0.025	0.072 ± 0.018	0.075 ± 0.041	0.171 ± 0.108	0.574 ± 1.595	0.068 ± 0.032	0.193 ± 0.108	0.072	–	0.069	–
PBC3	1.993 ± 0.193	2.375 ± 0.215	2.786 ± 0.354	2.007 ± 0.154	2.111 ± 0.369	2.127 ± 0.176	2.730 ± 0.463	1.971 ± 0.114	2.029 ± 0.322	$\mathbf{3.829 \pm 0.484}$	2.081	–	2.087	2.033
Retinopathy	0.628 ± 1.706	0.796 ± 0.108	$\mathbf{1.554 \pm 0.348}$	0.529 ± 0.297	0.155 ± 0.098	0.352 ± 0.184	0.820 ± 0.071	0.507 ± 0.241	0.617 ± 0.261	0.852 ± 0.202	0.131	–	0.492	–
WHAS500	2.284 ± 0.344	3.165 ± 0.491	3.354 ± 0.195	2.383 ± 0.168	2.221 ± 0.188	2.260 ± 0.094	3.447 ± 0.792	2.218 ± 0.077	2.387 ± 0.362	$\mathbf{4.183 \pm 0.777}$	–	–	2.272	2.578
GBSG2	0.748 ± 0.432	0.955 ± 0.212	1.540 ± 0.206	0.696 ± 0.144	0.636 ± 0.155	0.868 ± 0.408	1.240 ± 0.741	0.742 ± 0.116	0.692 ± 0.103	$\mathbf{2.470 \pm 0.172}$	0.802	–	0.684	–
Colon	0.337 ± 0.072	0.541 ± 0.086	1.519 ± 0.242	0.321 ± 0.021	0.328 ± 0.087	0.513 ± 0.233	1.220 ± 1.010	0.433 ± 0.005	0.341 ± 0.016	$\mathbf{1.772 \pm 0.360}$	1.417	–	0.398	–
GRACE	0.333 ± 0.034	0.446 ± 0.101	0.677 ± 0.548	0.316 ± 0.033	0.319 ± 0.018	0.395 ± 0.162	$\mathbf{1.192 \pm 2.913}$	0.603 ± 0.047	0.321 ± 0.026	0.666 ± 0.091	–	–	0.351	–
TRACE	0.082 ± 0.024	0.175 ± 0.129	$\mathbf{0.856 \pm 1.248}$	0.069 ± 0.012	0.074 ± 0.025	0.177 ± 0.106	0.238 ± 0.079	0.737 ± 1.768	0.108 ± 0.013	0.294 ± 0.008	–	–	0.130	–
Framingham	0.732 ± 0.013	0.932 ± 0.065	1.008 ± 0.220	0.743 ± 0.007	0.718 ± 0.036	0.780 ± 0.126	1.368 ± 0.122	1.252 ± 1.082	0.786 ± 0.089	$\mathbf{1.567 \pm 0.089}$	–	–	0.755	–
Blood Transfusion	0.492 ± 0.321	0.565 ± 0.155	0.533 ± 0.311	0.240 ± 0.049	0.207 ± 0.020	0.353 ± 0.125	0.569 ± 0.578	0.542 ± 0.148	0.276 ± 0.073	$\mathbf{0.975 \pm 0.200}$	–	–	0.322	–
QSAR Biodeg	3.585 ± 0.207	4.837 ± 0.670	3.695 ± 0.513	4.228 ± 0.410	$\mathbf{22.772 \pm 4.344}$	3.096 ± 0.058	6.437 ± 2.976	3.072 ± 0.274	3.949 ± 0.307	19.399 ± 7.340	–	–	3.468	–
Maternal Risk	0.621 ± 0.290	1.075 ± 0.224	1.418 ± 0.418	0.741 ± 0.130	0.739 ± 0.170	0.767 ± 0.248	1.263 ± 0.936	0.714 ± 0.079	0.651 ± 0.015	$\mathbf{1.582 \pm 0.269}$	0.585	–	0.610	0.775
Website Phishing	1.520 ± 0.456	1.805 ± 0.840	$\mathbf{2.000 \pm 0.000}$	1.805 ± 0.840	1.414 ± 0.000	1.414 ± 0.000	1.609 ± 0.840	1.414 ± 0.000	1.414 ± 0.000	1.414 ± 0.000	–	–	1.414	–

Table 14: NNDR (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.879 ± 0.087	0.918 ± 0.017	$\mathbf{0.983 \pm 0.016}$	0.880 ± 0.024	0.912	0.906 ± 0.089	0.930 ± 0.014	0.909 ± 0.067	0.891 ± 0.037	0.927 ± 0.061	0.875	0.880	0.918	0.900
Breast Cancer	$\mathbf{0.979 \pm 0.008}$	0.958 ± 0.022	0.965 ± 0.003	0.931 ± 0.012	0.929	0.949 ± 0.033	0.877 ± 0.241	0.975 ± 0.010	0.956 ± 0.004	0.972 ± 0.025	–	–	0.966	–
Liver Disorders	0.926 ± 0.035	0.877 ± 0.049	0.916 ± 0.077	0.834 ± 0.034	0.867	0.897 ± 0.053	0.860 ± 0.027	$\mathbf{1.000 \pm 0.000}$	0.856 ± 0.105	0.864 ± 0.059	0.878	–	0.859	–
Parkinsons	0.962 ± 0.011	0.956 ± 0.040	0.953 ± 0.019	0.907 ± 0.063	0.828	0.935 ± 0.012	0.944 ± 0.036	$\mathbf{0.992 \pm 0.016}$	0.935 ± 0.010	0.969 ± 0.022	–	–	0.948	–
Obesity	0.909 ± 0.024	0.929 ± 0.014	$\mathbf{0.963 \pm 0.010}$	0.925 ± 0.020	0.903	0.910 ± 0.019	0.935 ± 0.023	0.910 ± 0.014	0.923 ± 0.002	0.956 ± 0.013	–	–	0.923	–
Diabetes	0.835 ± 0.017	0.834 ± 0.010	0.897 ± 0.027	0.847 ± 0.012	–	0.846 ± 0.026	0.871 ± 0.034	0.848 ± 0.056	0.831 ± 0.013	0.905 ± 0.021	0.851	–	0.840	**0.928**
German Credit	0.962 ± 0.019	0.960 ± 0.001	$\mathbf{0.986 \pm 0.006}$	0.957 ± 0.017	–	0.951 ± 0.015	0.956 ± 0.007	0.949 ± 0.010	0.953 ± 0.017	0.962 ± 0.005	–	–	0.957	–
Bank Marketing	0.924 ± 0.005	0.938 ± 0.008	$\mathbf{0.966 \pm 0.010}$	0.945 ± 0.006	–	0.943 ± 0.003	0.955 ± 0.010	0.949 ± 0.018	0.945 ± 0.008	0.892 ± 0.164	0.944	–	0.929	–
Census Income	0.935 ± 0.003	0.944 ± 0.007	$\mathbf{0.970 \pm 0.009}$	0.948 ± 0.005	–	0.947 ± 0.005	0.962 ± 0.010	0.966 ± 0.025	0.952 ± 0.004	0.957 ± 0.028	0.941	–	0.930	–
Melanoma	0.870 ± 0.159	0.824 ± 0.076	0.922 ± 0.056	0.809 ± 0.051	0.807 ± 0.156	0.830 ± 0.053	0.832 ± 0.049	$\mathbf{1.000 \pm 0.000}$	0.805 ± 0.071	0.759 ± 0.029	0.855	–	0.805	–
PBC	0.708 ± 0.392	0.595 ± 0.172	$\mathbf{0.886 \pm 0.319}$	0.516 ± 0.074	0.498 ± 0.048	0.542 ± 0.145	0.638 ± 0.298	0.817 ± 0.409	0.538 ± 0.026	0.493 ± 0.096	0.563	–	0.665	–
PBC3	0.930 ± 0.042	0.916 ± 0.022	$\mathbf{0.970 \pm 0.008}$	0.899 ± 0.015	0.902 ± 0.006	0.911 ± 0.033	0.918 ± 0.024	0.940 ± 0.030	0.916 ± 0.034	0.932 ± 0.029	0.920	–	0.927	0.948
Retinopathy	0.692 ± 0.653	0.796 ± 0.041	0.979 ± 0.049	0.727 ± 0.120	0.441 ± 0.202	0.688 ± 0.059	0.810 ± 0.093	$\mathbf{1.000 \pm 0.000}$	0.680 ± 0.274	0.772 ± 0.036	0.500	–	0.756	–
WHAS500	0.925 ± 0.034	0.932 ± 0.027	$\mathbf{0.980 \pm 0.015}$	0.919 ± 0.010	0.895 ± 0.093	0.897 ± 0.072	0.926 ± 0.026	0.929 ± 0.013	0.907 ± 0.056	0.910 ± 0.073	–	–	0.903	0.946

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
GBSG2	0.883 ± 0.090	0.880 ± 0.013	0.971 ± 0.005	0.867 ± 0.044	0.853 ± 0.053	0.887 ± 0.038	0.888 ± 0.088	$\mathbf{0.973 \pm 0.055}$	0.850 ± 0.019	0.847 ± 0.087	0.901	–	0.857	–
Colon	0.724 ± 0.077	0.759 ± 0.044	0.985 ± 0.002	0.710 ± 0.098	0.688 ± 0.148	0.791 ± 0.091	0.841 ± 0.085	$\mathbf{1.000 \pm 0.000}$	0.703 ± 0.028	0.884 ± 0.055	0.976	–	0.805	–
GRACE	0.825 ± 0.046	0.806 ± 0.052	0.878 ± 0.104	0.762 ± 0.087	0.798 ± 0.026	0.824 ± 0.102	0.815 ± 0.068	$\mathbf{1.000 \pm 0.000}$	0.787 ± 0.066	0.789 ± 0.083	–	–	0.822	–
TRACE	0.705 ± 0.060	0.641 ± 0.101	0.894 ± 0.150	0.586 ± 0.060	0.616 ± 0.082	0.786 ± 0.114	0.727 ± 0.024	$\mathbf{0.980 \pm 0.086}$	0.734 ± 0.056	0.687 ± 0.056	–	–	0.771	–
Framingham	0.894 ± 0.010	0.892 ± 0.004	$\mathbf{0.924 \pm 0.007}$	0.888 ± 0.018	0.888 ± 0.012	0.898 ± 0.021	0.898 ± 0.028	0.919 ± 0.037	0.895 ± 0.004	0.883 ± 0.014	–	–	0.904	–
Blood Transfusion	0.972 ± 0.063	0.859 ± 0.114	0.891 ± 0.049	0.758 ± 0.150	0.795 ± 0.040	0.870 ± 0.109	0.846 ± 0.037	$\mathbf{1.000 \pm 0.000}$	0.823 ± 0.019	0.805 ± 0.133	–	–	0.861	–
QSAR Biodeg	0.968 ± 0.012	0.959 ± 0.021	$\mathbf{0.978 \pm 0.020}$	0.959 ± 0.019	0.972 ± 0.034	0.949 ± 0.009	0.970 ± 0.028	0.949 ± 0.006	0.961 ± 0.011	0.915 ± 0.284	–	–	0.952	–
Maternal Risk	0.976 ± 0.019	0.886 ± 0.059	0.975 ± 0.002	0.889 ± 0.035	0.876 ± 0.015	0.876 ± 0.082	0.889 ± 0.054	$\mathbf{1.000 \pm 0.000}$	0.861 ± 0.043	0.898 ± 0.031	0.864	–	0.884	0.903
Website Phishing	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	0.902 ± 0.420	$\mathbf{1.000 \pm 0.000}$	$\mathbf{1.000 \pm 0.000}$	–	–	**1.000**	–

Table 15: NNAA (0.5 ideal) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.591 ± 0.201	0.738 ± 0.052	0.967 ± 0.038	0.508 ± 0.131	0.655	0.600 ± 0.050	0.875 ± 0.151	0.629 ± 0.086	0.536 ± 0.023	0.927 ± 0.014	0.582	0.452	0.644	0.530
Breast Cancer	0.973 ± 0.019	0.958 ± 0.029	0.944 ± 0.017	0.639 ± 0.058	0.713	0.882 ± 0.055	0.996 ± 0.007	0.932 ± 0.030	0.753 ± 0.032	0.951 ± 0.003	–	–	0.913	–
Liver Disorders	0.629 ± 0.078	0.850 ± 0.037	0.673 ± 0.145	0.505 ± 0.080	0.458	0.711 ± 0.060	0.772 ± 0.160	0.905 ± 0.077	0.562 ± 0.070	0.891 ± 0.043	0.549	–	0.681	–
Parkinsons	0.858 ± 0.072	0.953 ± 0.017	0.934 ± 0.059	0.637 ± 0.053	0.638	0.907 ± 0.029	0.886 ± 0.122	0.936 ± 0.036	0.740 ± 0.072	0.893 ± 0.056	–	–	0.885	–
Obesity	0.672 ± 0.047	0.844 ± 0.027	0.935 ± 0.010	0.864 ± 0.088	0.726	0.775 ± 0.079	0.900 ± 0.109	0.761 ± 0.014	0.819 ± 0.062	0.950 ± 0.009	–	–	0.775	–
Diabetes	0.552 ± 0.019	0.628 ± 0.015	0.682 ± 0.042	0.691 ± 0.060	–	0.630 ± 0.065	0.828 ± 0.124	0.624 ± 0.155	0.636 ± 0.059	0.920 ± 0.013	0.697	–	0.575	0.813
German Credit	0.840 ± 0.231	0.703 ± 0.112	0.948 ± 0.003	0.614 ± 0.019	–	0.578 ± 0.048	0.775 ± 0.076	0.550 ± 0.011	0.686 ± 0.039	0.910 ± 0.014	–	–	0.611	–
Bank Marketing	0.585 ± 0.012	0.703 ± 0.034	0.806 ± 0.097	0.750 ± 0.033	–	0.775 ± 0.084	0.903 ± 0.087	0.784 ± 0.220	0.791 ± 0.040	0.995 ± 0.005	0.789	–	0.592	–
Census Income	0.688 ± 0.115	0.734 ± 0.034	0.789 ± 0.078	0.840 ± 0.163	–	0.828 ± 0.179	0.944 ± 0.119	0.908 ± 0.165	0.868 ± 0.029	0.990 ± 0.003	0.764	–	0.606	–
Melanoma	0.554 ± 0.146	0.744 ± 0.026	0.787 ± 0.169	0.576 ± 0.077	0.467 ± 0.083	0.641 ± 0.203	0.797 ± 0.225	0.884 ± 0.062	0.534 ± 0.160	0.763 ± 0.024	0.515	–	0.613	–
PBC	0.629 ± 0.763	0.566 ± 0.058	0.699 ± 0.045	0.500 ± 0.065	0.477 ± 0.076	0.562 ± 0.055	0.673 ± 0.064	0.802 ± 0.468	0.533 ± 0.055	0.701 ± 0.036	0.506	–	0.578	–
PBC3	0.550 ± 0.168	0.740 ± 0.084	0.906 ± 0.038	0.553 ± 0.069	0.596 ± 0.041	0.643 ± 0.049	0.880 ± 0.203	0.719 ± 0.035	0.624 ± 0.062	0.912 ± 0.005	0.670	–	0.631	0.720
Retinopathy	0.659 ± 0.708	0.797 ± 0.057	0.734 ± 0.097	0.635 ± 0.104	0.315 ± 0.035	0.550 ± 0.178	0.799 ± 0.084	0.691 ± 0.076	0.661 ± 0.089	0.811 ± 0.057	0.418	–	0.551	–
WHAS500	0.650 ± 0.013	0.808 ± 0.120	0.951 ± 0.027	0.533 ± 0.018	0.497 ± 0.111	0.586 ± 0.020	0.898 ± 0.169	0.654 ± 0.125	0.594 ± 0.080	0.912 ± 0.017	–	–	0.667	0.637
GBSG2	0.610 ± 0.338	0.778 ± 0.066	0.923 ± 0.033	0.579 ± 0.015	0.502 ± 0.001	0.750 ± 0.244	0.852 ± 0.164	0.749 ± 0.073	0.593 ± 0.071	0.959 ± 0.009	0.665	–	0.585	–
Colon	0.470 ± 0.070	0.649 ± 0.057	0.708 ± 0.102	0.491 ± 0.023	0.411 ± 0.020	0.699 ± 0.188	0.872 ± 0.220	0.762 ± 0.054	0.568 ± 0.083	0.875 ± 0.030	0.656	–	0.633	–
GRACE	0.531 ± 0.043	0.720 ± 0.079	0.797 ± 0.077	0.539 ± 0.035	0.507 ± 0.026	0.672 ± 0.141	0.821 ± 0.399	0.919 ± 0.027	0.536 ± 0.031	0.831 ± 0.029	–	–	0.624	–
TRACE	0.577 ± 0.053	0.696 ± 0.065	0.820 ± 0.145	0.537 ± 0.012	0.507 ± 0.009	0.761 ± 0.060	0.830 ± 0.033	0.946 ± 0.048	0.645 ± 0.028	0.830 ± 0.003	–	–	0.681	–
Framingham	0.518 ± 0.018	0.740 ± 0.063	0.817 ± 0.050	0.537 ± 0.032	0.495 ± 0.011	0.644 ± 0.165	0.936 ± 0.015	0.839 ± 0.402	0.636 ± 0.043	0.924 ± 0.013	–	–	0.619	–
Blood Transfusion	0.857 ± 0.271	0.900 ± 0.081	0.885 ± 0.065	0.627 ± 0.056	0.541 ± 0.066	0.814 ± 0.057	0.912 ± 0.133	0.975 ± 0.007	0.718 ± 0.076	0.956 ± 0.009	–	–	0.833	–
QSAR Biodeg	0.956 ± 0.026	0.946 ± 0.024	0.989 ± 0.009	0.967 ± 0.012	0.997 ± 0.004	0.890 ± 0.067	0.997 ± 0.002	0.897 ± 0.022	0.893 ± 0.024	0.995 ± 0.006	–	–	0.853	–
Maternal Risk	0.773 ± 0.057	0.882 ± 0.023	0.949 ± 0.012	0.807 ± 0.067	0.829 ± 0.058	0.860 ± 0.144	0.929 ± 0.089	0.890 ± 0.007	0.759 ± 0.016	0.911 ± 0.033	0.719	–	0.864	0.770
Website Phishing	0.368 ± 0.919	0.706 ± 0.071	0.875 ± 0.010	0.687 ± 0.053	0.022 ± 0.012	0.140 ± 0.047	0.520 ± 0.162	0.138 ± 0.026	0.174 ± 0.046	0.409 ± 0.049	–	–	0.120	–

Table 16: MIA AUROC (0.5 safe) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.494 ± 0.026	0.470 ± 0.077	0.477 ± 0.120	0.493 ± 0.061	0.486	0.515 ± 0.148	0.491 ± 0.105	0.458 ± 0.046	0.481 ± 0.049	0.492 ± 0.061	0.421	0.444	0.418	0.467
Breast Cancer	0.511 ± 0.085	0.500 ± 0.056	0.497 ± 0.059	0.497 ± 0.070	0.491	0.511 ± 0.140	0.521 ± 0.132	0.499 ± 0.101	0.496 ± 0.095	0.498 ± 0.064	–	–	0.488	–
Liver Disorders	0.540 ± 0.103	0.511 ± 0.085	0.500 ± 0.052	0.498 ± 0.078	0.503	0.525 ± 0.087	0.494 ± 0.049	0.514 ± 0.055	0.500 ± 0.063	0.508 ± 0.130	0.489	–	0.512	–
Parkinsons	0.511 ± 0.107	0.501 ± 0.058	0.483 ± 0.110	0.486 ± 0.071	0.510	0.499 ± 0.039	0.487 ± 0.073	0.536 ± 0.118	0.502 ± 0.076	0.536 ± 0.037	–	–	0.500	–
Obesity	0.544 ± 0.032	0.494 ± 0.001	0.503 ± 0.007	0.488 ± 0.036	0.527	0.515 ± 0.067	0.492 ± 0.051	0.512 ± 0.024	0.498 ± 0.035	0.499 ± 0.019	–	–	0.513	–
Diabetes	0.503 ± 0.011	0.508 ± 0.027	0.504 ± 0.014	0.497 ± 0.022	–	0.497 ± 0.010	0.498 ± 0.006	0.500 ± 0.012	0.490 ± 0.017	0.501 ± 0.022	0.502	–	0.502	0.500
German Credit	0.516 ± 0.029	0.520 ± 0.038	0.506 ± 0.061	0.522 ± 0.031	–	0.530 ± 0.026	0.497 ± 0.021	0.552 ± 0.028	0.511 ± 0.041	0.524 ± 0.044	–	–	0.527	–
Bank Marketing	0.503 ± 0.020	0.503 ± 0.034	0.506 ± 0.022	0.504 ± 0.026	–	0.498 ± 0.006	0.502 ± 0.008	0.501 ± 0.006	0.503 ± 0.024	0.498 ± 0.017	0.498	–	0.503	–
Census Income	0.503 ± 0.013	0.493 ± 0.015	0.501 ± 0.014	0.507 ± 0.020	–	0.507 ± 0.020	0.497 ± 0.013	0.507 ± 0.010	0.496 ± 0.028	0.500 ± 0.021	0.493	–	0.498	–
Melanoma	0.584 ± 0.076	0.572 ± 0.060	0.550 ± 0.076	0.526 ± 0.041	0.588 ± 0.064	0.538 ± 0.059	0.513 ± 0.057	0.563 ± 0.117	0.561 ± 0.084	0.486 ± 0.157	0.570	–	0.551	–
PBC	0.558 ± 0.154	0.510 ± 0.078	0.539 ± 0.110	0.529 ± 0.036	0.558 ± 0.020	0.522 ± 0.028	0.511 ± 0.101	0.525 ± 0.083	0.518 ± 0.061	0.508 ± 0.061	0.549	–	0.526	–
PBC3	0.559 ± 0.083	0.502 ± 0.094	0.490 ± 0.052	0.495 ± 0.022	0.602 ± 0.134	0.555 ± 0.113	0.496 ± 0.065	0.533 ± 0.082	0.507 ± 0.094	0.501 ± 0.106	0.520	–	0.573	0.495
Retinopathy	0.540 ± 0.094	0.502 ± 0.016	0.497 ± 0.047	0.509 ± 0.075	0.533 ± 0.018	0.500 ± 0.056	0.491 ± 0.088	0.535 ± 0.078	0.496 ± 0.079	0.499 ± 0.022	0.518	–	0.534	–
WHAS500	0.522 ± 0.081	0.523 ± 0.057	0.497 ± 0.051	0.538 ± 0.058	0.579 ± 0.068	0.519 ± 0.034	0.507 ± 0.038	0.525 ± 0.030	0.499 ± 0.086	0.484 ± 0.057	–	–	0.508	0.552
GBSG2	0.516 ± 0.073	0.482 ± 0.045	0.484 ± 0.079	0.499 ± 0.067	0.512 ± 0.105	0.501 ± 0.067	0.497 ± 0.078	0.495 ± 0.094	0.481 ± 0.064	0.481 ± 0.053	0.510	–	0.537	–
Colon	0.499 ± 0.010	0.491 ± 0.023	0.485 ± 0.030	0.466 ± 0.015	0.508 ± 0.046	0.489 ± 0.075	0.482 ± 0.039	0.478 ± 0.028	0.468 ± 0.011	0.516 ± 0.076	0.491	–	0.500	–
GRACE	0.533 ± 0.030	0.476 ± 0.011	0.497 ± 0.051	0.485 ± 0.044	0.524 ± 0.011	0.495 ± 0.064	0.514 ± 0.069	0.502 ± 0.057	0.489 ± 0.012	0.493 ± 0.006	–	–	0.526	–
TRACE	0.527 ± 0.018	0.496 ± 0.039	0.498 ± 0.005	0.503 ± 0.060	0.511 ± 0.046	0.498 ± 0.059	0.497 ± 0.010	0.499 ± 0.006	0.508 ± 0.028	0.497 ± 0.042	–	–	0.528	–
Framingham	0.519 ± 0.011	0.509 ± 0.022	0.508 ± 0.008	0.507 ± 0.026	0.517 ± 0.038	0.509 ± 0.014	0.505 ± 0.030	0.508 ± 0.024	0.509 ± 0.051	0.507 ± 0.057	–	–	0.507	–
Blood Transfusion	0.497 ± 0.040	0.511 ± 0.014	0.507 ± 0.058	0.493 ± 0.217	0.537 ± 0.023	0.505 ± 0.006	0.517 ± 0.058	0.512 ± 0.104	0.526 ± 0.041	0.499 ± 0.102	–	–	0.516	–
QSAR Biodeg	0.507 ± 0.020	0.490 ± 0.058	0.497 ± 0.034	0.514 ± 0.033	0.498 ± 0.051	0.511 ± 0.015	0.485 ± 0.048	0.507 ± 0.021	0.497 ± 0.051	0.484 ± 0.025	–	–	0.516	–
Maternal Risk	0.561 ± 0.039	0.508 ± 0.122	0.487 ± 0.010	0.470 ± 0.015	0.511 ± 0.119	0.501 ± 0.142	0.493 ± 0.096	0.502 ± 0.088	0.506 ± 0.112	0.500 ± 0.046	0.494	–	0.477	0.437
Website Phishing	0.530 ± 0.028	0.502 ± 0.063	0.530 ± 0.017	0.499 ± 0.051	0.708 ± 0.046	0.537 ± 0.073	0.498 ± 0.112	0.513 ± 0.041	0.520 ± 0.030	0.507 ± 0.019	–	–	0.555	–

Table 17: Duplicate rate (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	**0.000**	**0.000**	0.053
Breast Cancer	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	–	–	**0.000**	–
Liver Disorders	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	–	0.004	–
Parkinsons	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	–	–	**0.000**	–
Obesity	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	–	–	**0.000**	–
Diabetes	$\mathbf{0.000 \pm 0.000}$	0.000 ± 0.000	0.001 ± 0.001	$\mathbf{0.000 \pm 0.000}$	–	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	–	**0.000**	0.073
German Credit	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	–	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	–	–	**0.000**	–
Bank Marketing	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	–	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	**0.000**	–	**0.000**	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Census Income	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	**0.000**	–
Melanoma	**0.000 ± 0.000**	**0.000 ± 0.000**	0.002 ± 0.009	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	**0.000**	–
PBC	**0.000 ± 0.000**	0.011 ± 0.015	**0.000 ± 0.000**	**0.000 ± 0.000**	0.020 ± 0.000	**0.000 ± 0.000**	0.003 ± 0.006	0.017 ± 0.075	0.004 ± 0.010	**0.000 ± 0.000**	**0.000**	–	**0.000**	–
PBC3	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	**0.000**	0.040
Retinopathy	**0.000 ± 0.000**	0.011 ± 0.021	0.327 ± 0.140	0.056 ± 0.027	0.232 ± 0.020	0.119 ± 0.110	0.024 ± 0.022	0.101 ± 0.047	0.056 ± 0.055	0.008 ± 0.010	**0.000**	–	0.170	–
WHAS500	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	**0.000**	0.027
GBSG2	**0.000 ± 0.000**	**0.000 ± 0.000**	0.001 ± 0.003	**0.000 ± 0.000**	0.001 ± 0.003	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	**0.000**	–
Colon	**0.000 ± 0.000**	0.036 ± 0.020	0.330 ± 0.053	0.081 ± 0.019	0.087 ± 0.017	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	0.003	–
GRACE	**0.000 ± 0.000**	0.000 ± 0.002	0.005 ± 0.010	0.002 ± 0.006	0.006 ± 0.005	0.003 ± 0.003	0.001 ± 0.003	0.000 ± 0.002	0.003 ± 0.005	**0.000 ± 0.000**	–	–	0.005	–
TRACE	**0.000 ± 0.000**	0.000 ± 0.001	0.001 ± 0.002	0.001 ± 0.002	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	**0.000**	–
Framingham	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	**0.000**	–
Blood Transfusion	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.033 ± 0.042	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	**0.000**	–
QSAR Biodeg	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	**0.000**	–
Maternal Risk	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	**0.000**	0.007
Website Phishing	0.402 ± 0.866	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	0.796 ± 0.039	0.343 ± 0.046	0.157 ± 0.079	0.319 ± 0.044	0.298 ± 0.129	0.055 ± 0.005	–	–	0.591	–

Diversity / coverage (PRDC)

Table 18: Precision (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.980 ± 0.006	0.812 ± 0.120	**1.000 ± 0.000**	0.943 ± 0.043	0.477	0.967 ± 0.021	0.619 ± 0.360	0.996 ± 0.018	0.940 ± 0.032	0.457 ± 0.146	0.950	0.987	0.987	0.993
Breast Cancer	0.488 ± 0.175	0.004 ± 0.016	0.221 ± 0.149	**0.853 ± 0.090**	0.433	0.350 ± 0.035	0.001 ± 0.006	0.541 ± 0.033	0.035 ± 0.068	0.054 ± 0.121	–	–	0.429	–
Liver Disorders	0.994 ± 0.014	0.571 ± 0.315	**0.999 ± 0.005**	0.981 ± 0.005	0.978	0.982 ± 0.024	0.704 ± 0.095	0.998 ± 0.005	0.937 ± 0.050	0.594 ± 0.172	0.989	–	0.993	–
Parkinsons	0.812 ± 0.130	0.004 ± 0.018	0.553 ± 0.233	**0.848 ± 0.112**	0.814	0.372 ± 0.115	0.013 ± 0.016	0.842 ± 0.072	0.158 ± 0.113	0.107 ± 0.096	–	–	0.712	–
Obesity	**0.905 ± 0.060**	0.484 ± 0.015	0.874 ± 0.047	0.587 ± 0.042	0.688	0.684 ± 0.331	0.374 ± 0.191	0.702 ± 0.062	0.526 ± 0.095	0.098 ± 0.120	–	–	0.714	–
Diabetes	0.983 ± 0.003	0.940 ± 0.035	**0.995 ± 0.005**	0.874 ± 0.099	–	0.965 ± 0.032	0.757 ± 0.675	0.917 ± 0.214	0.916 ± 0.041	0.514 ± 0.068	0.981	–	0.979	0.980
German Credit	0.802 ± 0.501	0.734 ± 0.078	**1.000 ± 0.002**	0.822 ± 0.036	–	0.888 ± 0.016	0.700 ± 0.094	0.926 ± 0.022	0.671 ± 0.019	0.150 ± 0.053	–	–	0.953	–
Bank Marketing	0.958 ± 0.021	0.720 ± 0.082	**0.990 ± 0.009**	0.664 ± 0.165	–	0.743 ± 0.089	0.392 ± 0.130	0.761 ± 0.254	0.636 ± 0.049	0.323 ± 0.658	0.941	–	0.959	–
Census Income	0.940 ± 0.079	0.543 ± 0.086	**0.983 ± 0.012**	0.405 ± 0.417	–	0.480 ± 0.355	0.186 ± 0.298	0.377 ± 0.516	0.321 ± 0.141	0.017 ± 0.054	0.932	–	0.942	–
Melanoma	0.982 ± 0.040	0.807 ± 0.187	0.990 ± 0.009	0.961 ± 0.032	0.909 ± 0.055	**0.994 ± 0.026**	0.789 ± 0.091	0.986 ± 0.038	0.953 ± 0.068	0.831 ± 0.057	0.988	–	0.945	–
PBC	0.718 ± 1.060	0.912 ± 0.085	**1.000 ± 0.000**	0.981 ± 0.032	0.997 ± 0.006	0.991 ± 0.012	0.914 ± 0.168	0.888 ± 0.467	0.993 ± 0.021	0.841 ± 0.066	0.988	–	0.988	–
PBC3	0.984 ± 0.005	0.659 ± 0.163	**1.000 ± 0.000**	0.915 ± 0.014	0.568 ± 0.096	0.944 ± 0.094	0.456 ± 0.253	0.995 ± 0.014	0.906 ± 0.037	0.225 ± 0.181	0.953	–	0.982	**1.000**
Retinopathy	0.713 ± 0.769	0.476 ± 0.081	**0.994 ± 0.013**	0.655 ± 0.106	0.977 ± 0.035	0.815 ± 0.167	0.515 ± 0.182	0.737 ± 0.030	0.639 ± 0.156	0.442 ± 0.086	0.976	–	0.888	–
WHAS500	0.985 ± 0.019	0.596 ± 0.268	0.997 ± 0.007	0.954 ± 0.024	0.785 ± 0.058	0.960 ± 0.031	0.341 ± 0.392	**1.000 ± 0.000**	0.938 ± 0.054	0.725 ± 0.314	–	–	0.976	0.987
GBSG2	0.904 ± 0.335	0.648 ± 0.246	0.927 ± 0.172	0.917 ± 0.044	0.952 ± 0.068	0.937 ± 0.053	0.648 ± 0.300	0.916 ± 0.051	0.871 ± 0.072	0.540 ± 0.203	0.863	–	**0.974**	–
Colon	0.986 ± 0.032	0.903 ± 0.090	**0.996 ± 0.013**	0.934 ± 0.016	0.982 ± 0.013	0.976 ± 0.022	0.712 ± 0.280	0.961 ± 0.013	0.928 ± 0.083	0.593 ± 0.073	0.996	–	0.972	–
GRACE	0.992 ± 0.017	0.929 ± 0.038	0.997 ± 0.005	0.980 ± 0.018	0.975 ± 0.016	0.989 ± 0.025	0.560 ± 1.175	**1.000 ± 0.002**	0.988 ± 0.010	0.665 ± 0.016	–	–	0.996	–
TRACE	**0.998 ± 0.005**	0.834 ± 0.080	0.980 ± 0.024	0.970 ± 0.012	0.989 ± 0.003	0.995 ± 0.001	0.755 ± 0.208	0.899 ± 0.405	0.982 ± 0.004	0.641 ± 0.036	–	–	0.994	–
Framingham	0.968 ± 0.007	0.646 ± 0.137	**0.984 ± 0.013**	0.955 ± 0.001	0.970 ± 0.012	0.938 ± 0.080	0.301 ± 0.513	0.652 ± 1.006	0.904 ± 0.071	0.272 ± 0.078	–	–	0.970	–
Blood Transfusion	0.694 ± 0.557	0.345 ± 0.101	0.771 ± 0.088	0.947 ± 0.052	**0.987 ± 0.009**	0.829 ± 0.126	0.553 ± 0.401	0.581 ± 0.036	0.658 ± 0.207	0.347 ± 0.024	–	–	0.620	–
QSAR Biodeg	0.201 ± 0.063	0.052 ± 0.005	**0.419 ± 0.150**	0.093 ± 0.070	0.175 ± 0.154	0.251 ± 0.076	0.006 ± 0.010	0.348 ± 0.041	0.073 ± 0.031	0.040 ± 0.014	–	–	0.202	–
Maternal Risk	0.941 ± 0.029	0.620 ± 0.360	0.934 ± 0.179	0.781 ± 0.099	0.692 ± 0.151	0.757 ± 0.314	0.476 ± 0.331	0.838 ± 0.111	0.778 ± 0.134	0.778 ± 0.042	0.914	–	0.922	**0.953**
Website Phishing	0.990 ± 0.042	0.913 ± 0.024	0.998 ± 0.007	0.936 ± 0.032	**1.000 ± 0.000**	0.997 ± 0.007	0.969 ± 0.064	0.997 ± 0.007	0.992 ± 0.013	0.882 ± 0.071	–	–	**1.000**	–

Table 19: Recall (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.865 ± 0.357	0.886 ± 0.037	0.031 ± 0.069	0.918 ± 0.042	**0.979**	0.879 ± 0.055	0.907 ± 0.094	0.669 ± 0.154	0.923 ± 0.049	0.975 ± 0.091	0.887	0.954	0.816	0.858
Breast Cancer	0.100 ± 0.056	0.651 ± 0.274	0.286 ± 0.066	0.895 ± 0.068	**0.960**	0.738 ± 0.152	0.588 ± 0.549	0.330 ± 0.109	0.913 ± 0.023	0.525 ± 0.210	–	–	0.484	–
Liver Disorders	0.912 ± 0.058	0.891 ± 0.199	0.563 ± 0.307	0.972 ± 0.005	0.971	0.762 ± 0.185	0.960 ± 0.080	0.199 ± 0.293	0.969 ± 0.045	**1.000 ± 0.000**	0.877	–	0.877	–
Parkinsons	0.530 ± 0.170	0.667 ± 0.016	0.434 ± 0.199	0.981 ± 0.000	**0.987**	0.684 ± 0.189	0.838 ± 0.151	0.363 ± 0.161	0.944 ± 0.115	0.447 ± 0.160	–	–	0.699	–
Obesity	0.924 ± 0.016	0.915 ± 0.020	0.351 ± 0.032	0.854 ± 0.197	0.933	0.907 ± 0.011	0.810 ± 0.145	0.881 ± 0.026	0.907 ± 0.059	**0.949 ± 0.054**	–	–	0.871	–
Diabetes	0.947 ± 0.074	0.963 ± 0.025	0.704 ± 0.143	0.973 ± 0.022	–	0.940 ± 0.070	0.910 ± 0.152	0.946 ± 0.056	0.964 ± 0.023	**0.988 ± 0.019**	0.867	–	0.960	0.684
German Credit	0.420 ± 0.504	0.907 ± 0.079	0.015 ± 0.002	0.932 ± 0.022	–	0.924 ± 0.035	0.917 ± 0.097	0.875 ± 0.005	0.940 ± 0.078	**0.988 ± 0.025**	–	–	0.775	–
Bank Marketing	0.877 ± 0.081	0.936 ± 0.016	0.298 ± 0.184	0.814 ± 0.161	–	0.840 ± 0.125	0.871 ± 0.072	0.775 ± 0.157	0.892 ± 0.103	**0.949 ± 0.184**	0.593	–	0.839	–
Census Income	0.736 ± 0.056	0.911 ± 0.096	0.304 ± 0.172	0.883 ± 0.062	–	0.787 ± 0.036	0.631 ± 1.049	0.458 ± 0.739	0.840 ± 0.097	**0.917 ± 0.265**	0.673	–	0.774	–
Melanoma	0.967 ± 0.009	0.935 ± 0.159	0.423 ± 0.254	0.974 ± 0.023	0.990 ± 0.023	0.872 ± 0.215	0.929 ± 0.229	0.522 ± 0.273	0.986 ± 0.009	**1.000 ± 0.000**	0.927	–	0.933	–
PBC	0.807 ± 0.556	0.989 ± 0.030	0.553 ± 0.038	0.980 ± 0.010	0.993 ± 0.012	0.968 ± 0.043	0.968 ± 0.056	0.647 ± 0.709	0.972 ± 0.026	**1.000 ± 0.000**	0.948	–	0.912	–
PBC3	0.852 ± 0.145	0.910 ± 0.080	0.068 ± 0.039	0.946 ± 0.039	0.957 ± 0.058	0.842 ± 0.098	0.858 ± 0.298	0.455 ± 0.099	0.914 ± 0.041	**0.986 ± 0.032**	0.749	–	0.695	0.771
Retinopathy	0.668 ± 1.291	0.976 ± 0.044	0.244 ± 0.095	0.988 ± 0.034	**0.998 ± 0.005**	0.929 ± 0.022	0.965 ± 0.065	0.982 ± 0.042	0.976 ± 0.055	0.997 ± 0.015	0.969	–	0.942	–
WHAS500	0.735 ± 0.078	0.856 ± 0.263	0.047 ± 0.050	0.947 ± 0.014	0.957 ± 0.014	0.885 ± 0.110	0.906 ± 0.135	0.593 ± 0.238	0.925 ± 0.045	**1.000 ± 0.000**	–	–	0.742	0.851
GBSG2	0.925 ± 0.056	0.923 ± 0.098	0.190 ± 0.072	0.973 ± 0.014	0.957 ± 0.014	0.853 ± 0.142	0.904 ± 0.327	0.763 ± 0.053	0.964 ± 0.022	**0.985 ± 0.032**	0.880	–	0.916	–
Colon	0.964 ± 0.020	0.966 ± 0.040	0.259 ± 0.139	0.979 ± 0.013	0.983 ± 0.010	0.911 ± 0.139	0.788 ± 0.419	0.743 ± 0.028	0.978 ± 0.019	**0.995 ± 0.017**	0.500	–	0.929	–
GRACE	0.981 ± 0.011	0.965 ± 0.047	0.593 ± 0.454	0.982 ± 0.008	0.987 ± 0.016	0.811 ± 0.330	0.893 ± 0.419	0.299 ± 0.225	0.982 ± 0.006	**0.998 ± 0.002**	–	–	0.891	–
TRACE	0.957 ± 0.046	0.989 ± 0.003	0.476 ± 0.564	0.993 ± 0.008	0.990 ± 0.001	0.838 ± 0.191	0.977 ± 0.041	0.316 ± 0.097	0.993 ± 0.005	**1.000 ± 0.001**	–	–	0.921	–
Framingham	0.965 ± 0.013	0.974 ± 0.042	0.525 ± 0.025	0.970 ± 0.003	0.972 ± 0.008	0.904 ± 0.095	0.869 ± 0.253	0.663 ± 0.764	0.951 ± 0.026	**0.998 ± 0.006**	–	–	0.911	–
Blood Transfusion	0.611 ± 0.910	0.928 ± 0.120	0.583 ± 0.288	**0.995 ± 0.009**	0.992 ± 0.015	0.825 ± 0.166	0.928 ± 0.171	0.244 ± 0.098	0.980 ± 0.045	0.978 ± 0.055	–	–	0.946	–
QSAR Biodeg	0.547 ± 0.182	0.776 ± 0.066	0.141 ± 0.149	0.689 ± 0.111	0.805 ± 0.729	0.801 ± 0.106	0.119 ± 0.281	0.640 ± 0.014	0.772 ± 0.120	0.472 ± 1.128	–	–	**0.822**	–
Maternal Risk	0.873 ± 0.123	0.825 ± 0.298	0.154 ± 0.029	0.901 ± 0.196	0.962 ± 0.062	0.809 ± 0.260	0.807 ± 0.332	0.700 ± 0.072	0.953 ± 0.072	**0.972 ± 0.095**	0.909	–	0.878	0.842
Website Phishing	0.834 ± 0.687	0.836 ± 0.082	0.195 ± 0.103	0.826 ± 0.017	0.999 ± 0.002	**1.000 ± 0.000**	0.957 ± 0.168	**1.000 ± 0.000**	0.999 ± 0.005	**1.000 ± 0.000**	–	–	0.998	–

Table 20: Density (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	1.033 ± 0.080	0.489 ± 0.259	1.616 ± 0.212	0.904 ± 0.147	0.574	1.036 ± 0.204	0.250 ± 0.195	**1.623 ± 0.132**	0.942 ± 0.128	0.139 ± 0.040	1.092	1.152	1.362	1.381
Breast Cancer	0.249 ± 0.164	0.001 ± 0.003	0.078 ± 0.068	**0.590 ± 0.106**	0.507	0.135 ± 0.012	0.000 ± 0.001	0.556 ± 0.164	0.008 ± 0.016	0.011 ± 0.025	–	–	0.258	–
Liver Disorders	1.173 ± 0.078	0.268 ± 0.231	**1.375 ± 0.251**	0.946 ± 0.069	1.162	1.090 ± 0.271	0.370 ± 0.012	1.100 ± 0.161	0.874 ± 0.274	0.258 ± 0.097	1.284	–	1.243	–
Parkinsons	0.615 ± 0.063	0.001 ± 0.006	0.276 ± 0.178	0.575 ± 0.217	0.772	0.166 ± 0.143	0.003 ± 0.005	**0.992 ± 0.094**	0.054 ± 0.058	0.027 ± 0.032	–	–	0.427	–
Obesity	**0.815 ± 0.096**	0.225 ± 0.040	0.808 ± 0.126	0.286 ± 0.046	0.565	0.430 ± 0.399	0.153 ± 0.082	0.458 ± 0.060	0.252 ± 0.096	0.023 ± 0.027	–	–	0.461	–
Diabetes	0.994 ± 0.057	0.870 ± 0.156	**1.140 ± 0.043**	0.641 ± 0.120	–	0.941 ± 0.063	0.619 ± 0.783	0.836 ± 0.416	0.815 ± 0.133	0.213 ± 0.077	1.045	–	1.021	1.015
German Credit	0.554 ± 0.700	0.439 ± 0.132	**2.157 ± 0.320**	0.558 ± 0.014	–	0.784 ± 0.114	0.393 ± 0.095	0.911 ± 0.210	0.376 ± 0.051	0.039 ± 0.016	–	–	1.181	–
Bank Marketing	1.025 ± 0.232	0.449 ± 0.112	**1.565 ± 0.044**	0.347 ± 0.157	–	0.447 ± 0.163	0.147 ± 0.076	0.501 ± 0.476	0.319 ± 0.041	0.076 ± 0.150	0.849	–	1.113	–
Census Income	0.996 ± 0.553	0.328 ± 0.169	**1.667 ± 0.184**	0.156 ± 0.170	–	0.281 ± 0.381	0.055 ± 0.102	0.187 ± 0.351	0.132 ± 0.077	0.004 ± 0.011	0.921	–	1.239	–
Melanoma	1.133 ± 0.157	0.541 ± 0.088	1.111 ± 0.162	0.880 ± 0.170	0.981 ± 0.144	1.102 ± 0.087	0.512 ± 0.121	**1.292 ± 0.174**	0.923 ± 0.179	0.504 ± 0.131	1.074	–	0.999	–
PBC	0.703 ± 1.328	0.778 ± 0.173	**1.124 ± 0.094**	0.961 ± 0.102	1.056 ± 0.085	1.007 ± 0.066	0.779 ± 0.305	0.867 ± 1.303	1.016 ± 0.063	0.592 ± 0.136	1.035	–	1.099	–
PBC3	1.167 ± 0.075	0.344 ± 0.208	1.602 ± 0.237	0.746 ± 0.065	0.707 ± 0.183	0.881 ± 0.056	0.187 ± 0.100	**1.736 ± 0.377**	0.853 ± 0.235	0.066 ± 0.052	1.117	–	1.321	1.373
Retinopathy	0.674 ± 1.255	0.308 ± 0.036	**1.197 ± 0.106**	0.525 ± 0.122	1.112 ± 0.050	0.770 ± 0.443	0.341 ± 0.138	0.680 ± 0.047	0.508 ± 0.255	0.281 ± 0.122	1.002	–	0.899	–
WHAS500	1.342 ± 0.127	0.300 ± 0.216	1.686 ± 0.651	0.907 ± 0.141	1.010 ± 0.071	1.035 ± 0.207	0.152 ± 0.237	**1.693 ± 0.512**	0.889 ± 0.054	0.357 ± 0.281	–	–	1.318	1.305
GBSG2	0.851 ± 0.648	0.384 ± 0.225	1.146 ± 0.411	0.785 ± 0.085	0.995 ± 0.100	0.827 ± 0.514	0.304 ± 0.173	0.923 ± 0.108	0.728 ± 0.042	0.168 ± 0.090	0.869	–	**1.159**	–
Colon	1.090 ± 0.061	0.684 ± 0.087	1.145 ± 0.059	0.901 ± 0.013	1.060 ± 0.025	0.960 ± 0.116	0.372 ± 0.570	1.081 ± 0.013	0.893 ± 0.279	0.285 ± 0.047	**1.158**	–	1.139	–
GRACE	1.037 ± 0.048	0.693 ± 0.054	1.087 ± 0.099	0.951 ± 0.030	0.993 ± 0.052	1.012 ± 0.083	0.376 ± 0.806	**1.108 ± 0.088**	1.013 ± 0.043	0.360 ± 0.071	–	–	1.095	–
TRACE	1.073 ± 0.057	0.643 ± 0.118	0.982 ± 0.090	0.931 ± 0.047	1.024 ± 0.036	**1.082 ± 0.115**	0.525 ± 0.228	0.898 ± 0.838	1.023 ± 0.058	0.397 ± 0.022	–	–	1.067	–
Framingham	0.975 ± 0.062	0.391 ± 0.166	**1.136 ± 0.102**	0.871 ± 0.015	1.000 ± 0.007	0.823 ± 0.291	0.141 ± 0.272	0.498 ± 1.113	0.782 ± 0.154	0.098 ± 0.038	–	–	1.046	–
Blood Transfusion	0.569 ± 0.587	0.193 ± 0.057	0.687 ± 0.016	0.819 ± 0.157	**0.968 ± 0.077**	0.758 ± 0.137	0.338 ± 0.613	0.556 ± 0.052	0.574 ± 0.228	0.119 ± 0.043	–	–	0.500	–
QSAR Biodeg	0.062 ± 0.029	0.021 ± 0.003	**0.137 ± 0.053**	0.029 ± 0.017	0.035 ± 0.031	0.091 ± 0.030	0.001 ± 0.003	0.134 ± 0.031	0.021 ± 0.016	0.008 ± 0.003	–	–	0.066	–
Maternal Risk	**1.033 ± 0.245**	0.325 ± 0.269	0.984 ± 0.198	0.490 ± 0.104	0.403 ± 0.147	0.603 ± 0.727	0.238 ± 0.222	0.994 ± 0.262	0.591 ± 0.157	0.218 ± 0.034	0.956	–	0.999	0.860
Website Phishing	2.197 ± 2.674	0.839 ± 0.069	1.369 ± 0.228	0.884 ± 0.037	2.753 ± 0.046	2.272 ± 0.372	1.588 ± 0.531	2.224 ± 0.125	2.064 ± 0.492	0.979 ± 0.018	–	–	**2.918**	–

Table 21: Coverage (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.879 ± 0.202	0.681 ± 0.111	0.202 ± 0.097	0.968 ± 0.026	0.933	0.925 ± 0.027	0.400 ± 0.510	0.940 ± 0.026	0.961 ± 0.033	0.096 ± 0.000	0.941	**0.987**	0.954	0.912
Breast Cancer	0.222 ± 0.048	0.001 ± 0.003	0.144 ± 0.082	**0.789 ± 0.055**	0.778	0.304 ± 0.087	0.001 ± 0.003	0.305 ± 0.036	0.017 ± 0.025	0.008 ± 0.003	–	–	0.209	–
Liver Disorders	0.966 ± 0.044	0.349 ± 0.086	0.686 ± 0.218	0.961 ± 0.010	**0.989**	0.798 ± 0.090	0.566 ± 0.207	0.460 ± 0.289	0.929 ± 0.105	0.165 ± 0.034	0.913	–	0.855	–
Parkinsons	0.598 ± 0.160	0.004 ± 0.018	0.267 ± 0.079	0.812 ± 0.144	**0.872**	0.318 ± 0.199	0.015 ± 0.024	0.476 ± 0.133	0.171 ± 0.187	0.021 ± 0.009	–	–	0.417	–
Obesity	**0.830 ± 0.103**	0.304 ± 0.070	0.283 ± 0.014	0.295 ± 0.169	0.725	0.540 ± 0.233	0.221 ± 0.115	0.561 ± 0.031	0.354 ± 0.048	0.028 ± 0.004	–	–	0.575	–
Diabetes	0.909 ± 0.114	0.852 ± 0.024	0.763 ± 0.123	0.726 ± 0.154	–	0.902 ± 0.061	0.529 ± 0.218	0.861 ± 0.189	0.833 ± 0.081	0.164 ± 0.022	0.647	–	**0.949**	0.156
German Credit	0.590 ± 0.897	0.674 ± 0.127	0.310 ± 0.043	0.826 ± 0.006	–	0.905 ± 0.039	0.522 ± 0.128	0.909 ± 0.050	0.670 ± 0.065	0.076 ± 0.018	–	–	**0.949**	–
Bank Marketing	0.932 ± 0.012	0.693 ± 0.099	0.712 ± 0.146	0.581 ± 0.015	–	0.616 ± 0.163	0.237 ± 0.197	0.632 ± 0.369	0.502 ± 0.087	0.006 ± 0.008	0.577	–	**0.945**	–
Census Income	0.861 ± 0.083	0.527 ± 0.060	0.703 ± 0.154	0.288 ± 0.317	–	0.474 ± 0.425	0.100 ± 0.186	0.327 ± 0.512	0.306 ± 0.164	0.003 ± 0.000	0.635	–	**0.929**	–
Melanoma	0.959 ± 0.086	0.673 ± 0.106	0.453 ± 0.280	0.923 ± 0.032	0.972 ± 0.057	0.850 ± 0.229	0.516 ± 0.441	0.673 ± 0.166	0.931 ± 0.168	0.557 ± 0.155	**0.976**	–	0.835	–
PBC	0.481 ± 1.052	0.912 ± 0.098	0.562 ± 0.035	0.965 ± 0.025	**0.972 ± 0.010**	0.896 ± 0.095	0.778 ± 0.075	0.554 ± 0.781	0.940 ± 0.020	0.708 ± 0.120	0.964	–	0.900	–
PBC3	**0.949 ± 0.072**	0.596 ± 0.054	0.388 ± 0.063	0.897 ± 0.042	0.944 ± 0.058	0.872 ± 0.029	0.331 ± 0.294	0.888 ± 0.005	0.889 ± 0.079	0.082 ± 0.031	0.849	–	0.943	0.746
Retinopathy	0.685 ± 1.133	0.651 ± 0.123	0.308 ± 0.066	0.881 ± 0.034	**0.991 ± 0.020**	0.867 ± 0.250	0.591 ± 0.117	0.917 ± 0.020	0.773 ± 0.148	0.612 ± 0.228	0.980	–	0.912	–
WHAS500	0.921 ± 0.047	0.438 ± 0.301	0.271 ± 0.095	0.935 ± 0.036	**0.984 ± 0.018**	0.936 ± 0.016	0.281 ± 0.290	0.914 ± 0.068	0.894 ± 0.007	0.113 ± 0.044	–	–	0.959	0.812
GBSG2	0.858 ± 0.306	0.571 ± 0.188	0.220 ± 0.060	0.915 ± 0.025	**0.962 ± 0.015**	0.691 ± 0.396	0.411 ± 0.316	0.804 ± 0.046	0.881 ± 0.121	0.069 ± 0.016	0.743	–	0.945	–
Colon	0.959 ± 0.078	0.795 ± 0.129	0.321 ± 0.204	0.943 ± 0.036	**0.983 ± 0.020**	0.794 ± 0.301	0.399 ± 0.498	0.817 ± 0.036	0.935 ± 0.025	0.255 ± 0.073	0.494	–	0.936	–
GRACE	0.956 ± 0.042	0.662 ± 0.109	0.447 ± 0.174	0.954 ± 0.016	**0.970 ± 0.008**	0.739 ± 0.269	0.464 ± 0.996	0.481 ± 0.041	0.942 ± 0.035	0.447 ± 0.081	–	–	0.907	–
TRACE	0.926 ± 0.085	0.711 ± 0.105	0.314 ± 0.308	0.941 ± 0.033	**0.973 ± 0.017**	0.673 ± 0.156	0.519 ± 0.165	0.297 ± 0.427	0.890 ± 0.048	0.463 ± 0.053	–	–	0.877	–
Framingham	0.962 ± 0.024	0.611 ± 0.200	0.553 ± 0.144	0.942 ± 0.021	**0.970 ± 0.007**	0.844 ± 0.185	0.202 ± 0.072	0.405 ± 0.946	0.828 ± 0.036	0.138 ± 0.027	–	–	0.922	–
Blood Transfusion	0.368 ± 0.350	0.272 ± 0.172	0.347 ± 0.206	0.888 ± 0.059	**0.957 ± 0.079**	0.592 ± 0.222	0.282 ± 0.426	0.251 ± 0.157	0.785 ± 0.141	0.104 ± 0.022	–	–	0.730	–
QSAR Biodeg	0.126 ± 0.050	0.023 ± 0.006	0.086 ± 0.021	0.053 ± 0.027	0.001 ± 0.000	0.212 ± 0.065	0.003 ± 0.005	**0.248 ± 0.042**	0.048 ± 0.032	0.001 ± 0.000	–	–	0.145	–
Maternal Risk	**0.861 ± 0.144**	0.335 ± 0.079	0.245 ± 0.057	0.534 ± 0.112	0.675 ± 0.204	0.597 ± 0.484	0.270 ± 0.409	0.661 ± 0.068	0.715 ± 0.113	0.173 ± 0.046	0.817	–	0.762	0.501
Website Phishing	0.883 ± 0.440	0.751 ± 0.028	0.500 ± 0.007	0.788 ± 0.086	**1.000 ± 0.000**	0.988 ± 0.005	0.752 ± 0.176	0.982 ± 0.020	0.983 ± 0.013	0.838 ± 0.031	–	–	0.998	–

Detection / distribution

Table 22: C2ST AUROC (↓, 0.5 best) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.746 ± 0.240	0.956 ± 0.039	0.999 ± 0.003	0.902 ± 0.033	0.958	0.956 ± 0.014	0.976 ± 0.012	0.999 ± 0.003	0.939 ± 0.014	0.987 ± 0.003	0.860	**0.442**	0.948	0.830
Breast Cancer	1.000 ± 0.000	1.000 ± 0.000	0.993 ± 0.005	**0.817 ± 0.265**	0.828	0.989 ± 0.014	0.998 ± 0.001	1.000 ± 0.000	0.988 ± 0.008	0.998 ± 0.002	–	–	0.998	–
Liver Disorders	0.650 ± 0.131	0.960 ± 0.054	0.822 ± 0.062	0.593 ± 0.034	**0.461**	0.879 ± 0.055	0.914 ± 0.088	1.000 ± 0.000	0.648 ± 0.074	0.972 ± 0.002	0.578	–	0.823	–
Parkinsons	0.999 ± 0.005	0.995 ± 0.016	0.984 ± 0.012	0.732 ± 0.150	**0.728**	0.976 ± 0.021	0.993 ± 0.006	0.993 ± 0.016	0.951 ± 0.043	0.983 ± 0.010	–	–	0.995	–
Obesity	**0.797 ± 0.090**	0.993 ± 0.002	0.999 ± 0.001	0.991 ± 0.008	0.839	0.994 ± 0.003	0.997 ± 0.008	1.000 ± 0.000	0.993 ± 0.002	1.000 ± 0.000	–	–	0.997	–
Diabetes	**0.807 ± 0.057**	0.951 ± 0.022	0.927 ± 0.011	0.984 ± 0.002	–	0.999 ± 0.001	0.997 ± 0.002	1.000 ± 0.000	0.997 ± 0.002	0.997 ± 0.002	0.856	–	0.998	0.928
German Credit	1.000 ± 0.000	0.937 ± 0.035	0.999 ± 0.000	**0.877 ± 0.013**	–	0.892 ± 0.078	0.977 ± 0.013	1.000 ± 0.001	0.924 ± 0.011	0.996 ± 0.002	–	–	0.882	–
Bank Marketing	**0.833 ± 0.119**	0.910 ± 0.025	0.964 ± 0.023	0.985 ± 0.026	–	0.969 ± 0.062	0.998 ± 0.005	1.000 ± 0.000	0.968 ± 0.020	1.000 ± 0.000	0.965	–	0.901	–
Census Income	0.937 ± 0.166	0.965 ± 0.020	0.980 ± 0.014	0.990 ± 0.034	–	0.995 ± 0.004	0.998 ± 0.002	1.000 ± 0.001	0.993 ± 0.001	1.000 ± 0.000	**0.916**	–	0.994	–
Melanoma	0.686 ± 0.341	0.884 ± 0.098	0.924 ± 0.024	0.691 ± 0.121	0.726 ± 0.076	0.841 ± 0.123	0.922 ± 0.147	0.999 ± 0.002	0.717 ± 0.075	0.890 ± 0.077	**0.456**	–	0.761	–
PBC	0.700 ± 0.633	0.629 ± 0.084	0.839 ± 0.092	0.499 ± 0.148	**0.494 ± 0.017**	0.633 ± 0.068	0.790 ± 0.093	0.875 ± 0.458	0.536 ± 0.115	0.832 ± 0.045	0.563	–	0.653	–
PBC3	**0.751 ± 0.249**	0.944 ± 0.070	0.987 ± 0.002	0.802 ± 0.076	0.892 ± 0.009	0.912 ± 0.017	0.970 ± 0.064	1.000 ± 0.000	0.849 ± 0.014	0.986 ± 0.001	0.802	–	0.925	0.894
Retinopathy	0.820 ± 0.403	0.957 ± 0.039	0.939 ± 0.054	0.831 ± 0.063	**0.494 ± 0.061**	0.795 ± 0.202	0.948 ± 0.023	0.981 ± 0.014	0.881 ± 0.065	0.944 ± 0.012	0.526	–	0.819	–
WHAS500	1.000 ± 0.000	0.960 ± 0.076	0.991 ± 0.008	**0.568 ± 0.063**	0.713 ± 0.100	0.785 ± 0.049	0.971 ± 0.066	1.000 ± 0.000	0.675 ± 0.032	0.985 ± 0.004	–	–	0.953	0.669
GBSG2	0.756 ± 0.498	0.941 ± 0.051	0.987 ± 0.008	0.788 ± 0.025	**0.675 ± 0.027**	0.914 ± 0.151	0.959 ± 0.066	1.000 ± 0.000	0.767 ± 0.045	0.994 ± 0.002	0.845	–	0.837	–
Colon	0.672 ± 0.147	0.854 ± 0.053	0.943 ± 0.060	**0.652 ± 0.037**	0.776 ± 0.008	0.994 ± 0.003	0.997 ± 0.004	1.000 ± 0.000	0.995 ± 0.004	0.989 ± 0.005	0.859	–	0.977	–
GRACE	0.706 ± 0.117	0.885 ± 0.055	0.927 ± 0.042	0.697 ± 0.027	**0.618 ± 0.104**	0.867 ± 0.080	0.927 ± 0.167	1.000 ± 0.000	0.734 ± 0.040	0.942 ± 0.018	–	–	0.819	–
TRACE	0.643 ± 0.117	0.851 ± 0.083	0.926 ± 0.083	**0.614 ± 0.021**	0.664 ± 0.045	0.995 ± 0.003	0.992 ± 0.002	0.999 ± 0.003	0.993 ± 0.003	0.985 ± 0.004	–	–	0.978	–
Framingham	0.776 ± 0.012	0.914 ± 0.062	0.952 ± 0.015	0.771 ± 0.011	**0.650 ± 0.042**	0.910 ± 0.081	0.988 ± 0.003	0.996 ± 0.012	0.864 ± 0.031	0.988 ± 0.004	–	–	0.850	–
Blood Transfusion	0.984 ± 0.043	0.989 ± 0.017	0.996 ± 0.001	0.969 ± 0.030	**0.933 ± 0.063**	0.986 ± 0.010	0.989 ± 0.014	1.000 ± 0.000	0.979 ± 0.016	0.992 ± 0.005	–	–	0.982	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
QSAR Biodeg	0.999 ± 0.003	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	$\mathbf{0.999 \pm 0.004}$	0.999 ± 0.001	0.999 ± 0.003	1.000 ± 0.000	0.999 ± 0.001	1.000 ± 0.000	–	–	1.000	–
Maternal Risk	0.803 ± 0.043	0.998 ± 0.001	0.997 ± 0.002	0.992 ± 0.004	0.977 ± 0.020	0.993 ± 0.004	0.995 ± 0.008	0.999 ± 0.004	0.983 ± 0.002	0.995 ± 0.003	**0.706**	–	0.990	0.891
Website Phishing	0.775 ± 0.489	1.000 ± 0.000	1.000 ± 0.000	1.000 ± 0.000	$\mathbf{0.363 \pm 0.059}$	0.635 ± 0.052	0.927 ± 0.034	0.664 ± 0.047	0.707 ± 0.081	0.904 ± 0.029	–	–	0.564	–

Table 23: MMD^2 (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.035 ± 0.046	0.099 ± 0.049	0.392 ± 0.088	$\mathbf{0.000 \pm 0.000}$	0.188	0.033 ± 0.010	0.242 ± 0.210	0.027 ± 0.007	0.007 ± 0.011	0.399 ± 0.025	0.011	0.000	0.014	0.015
Breast Cancer	0.054 ± 0.013	0.218 ± 0.041	0.110 ± 0.068	$\mathbf{0.009 \pm 0.003}$	0.227	0.058 ± 0.013	0.806 ± 0.460	0.101 ± 0.028	0.039 ± 0.057	0.573 ± 0.050	–	–	0.056	–
Liver Disorders	0.002 ± 0.005	0.276 ± 0.126	0.118 ± 0.148	0.001 ± 0.002	**0.000**	0.066 ± 0.048	0.132 ± 0.156	0.066 ± 0.030	0.021 ± 0.034	0.480 ± 0.041	0.023	–	0.090	–
Parkinsons	0.027 ± 0.012	0.202 ± 0.086	0.140 ± 0.050	$\mathbf{0.000 \pm 0.000}$	0.121	0.096 ± 0.059	0.145 ± 0.265	0.119 ± 0.032	0.022 ± 0.016	0.488 ± 0.007	–	–	0.057	–
Obesity	$\mathbf{0.003 \pm 0.003}$	0.037 ± 0.027	0.081 ± 0.022	0.031 ± 0.042	0.014	0.008 ± 0.020	0.109 ± 0.200	0.006 ± 0.002	0.019 ± 0.010	0.202 ± 0.030	–	–	0.003	–
Diabetes	0.008 ± 0.013	0.019 ± 0.007	0.017 ± 0.019	0.019 ± 0.030	–	0.007 ± 0.007	0.157 ± 0.159	0.025 ± 0.086	0.019 ± 0.019	0.381 ± 0.085	0.060	–	**0.002**	0.104
German Credit	0.089 ± 0.209	0.024 ± 0.030	0.299 ± 0.013	$\mathbf{0.001 \pm 0.003}$	–	0.008 ± 0.009	0.049 ± 0.039	0.007 ± 0.004	0.015 ± 0.010	0.238 ± 0.022	–	–	0.002	–
Bank Marketing	0.005 ± 0.001	0.010 ± 0.006	0.033 ± 0.014	0.012 ± 0.009	–	0.021 ± 0.015	0.090 ± 0.130	0.019 ± 0.017	0.027 ± 0.018	0.873 ± 0.104	0.063	–	**0.002**	–
Census Income	0.020 ± 0.019	0.012 ± 0.010	0.039 ± 0.040	0.123 ± 0.294	–	0.048 ± 0.057	0.224 ± 0.647	0.065 ± 0.111	0.044 ± 0.032	0.676 ± 0.035	0.069	–	**0.003**	–
Melanoma	0.021 ± 0.081	0.153 ± 0.073	0.199 ± 0.186	0.005 ± 0.014	$\mathbf{0.001 \pm 0.002}$	0.036 ± 0.064	0.223 ± 0.252	0.031 ± 0.043	0.008 ± 0.013	0.179 ± 0.051	0.005	–	0.012	–
PBC	0.045 ± 0.089	0.028 ± 0.044	0.139 ± 0.084	0.004 ± 0.013	$\mathbf{0.000 \pm 0.000}$	0.029 ± 0.032	0.106 ± 0.069	0.109 ± 0.386	0.024 ± 0.023	0.153 ± 0.025	0.004	–	0.015	–
PBC3	0.019 ± 0.030	0.069 ± 0.053	0.194 ± 0.066	$\mathbf{0.000 \pm 0.001}$	0.107 ± 0.070	0.040 ± 0.015	0.225 ± 0.202	0.049 ± 0.023	0.016 ± 0.009	0.340 ± 0.021	0.027	–	0.017	0.033
Retinopathy	0.042 ± 0.130	0.029 ± 0.062	0.171 ± 0.060	0.006 ± 0.005	$\mathbf{0.000 \pm 0.000}$	0.018 ± 0.035	0.052 ± 0.049	0.007 ± 0.006	0.028 ± 0.019	0.045 ± 0.019	**0.000**	–	0.005	–
WHAS500	0.019 ± 0.039	0.115 ± 0.092	0.273 ± 0.110	$\mathbf{0.000 \pm 0.001}$	0.022 ± 0.026	0.017 ± 0.021	0.259 ± 0.275	0.034 ± 0.043	0.007 ± 0.003	0.394 ± 0.016	–	–	0.013	0.006
GBSG2	0.015 ± 0.033	0.107 ± 0.116	0.325 ± 0.082	0.007 ± 0.010	$\mathbf{0.002 \pm 0.002}$	0.087 ± 0.190	0.200 ± 0.235	0.020 ± 0.008	0.006 ± 0.008	0.661 ± 0.073	0.039	–	0.005	–
Colon	0.008 ± 0.013	0.035 ± 0.044	0.190 ± 0.145	0.001 ± 0.002	$\mathbf{0.000 \pm 0.000}$	0.052 ± 0.097	0.187 ± 0.228	0.009 ± 0.010	0.007 ± 0.005	0.386 ± 0.055	0.079	–	0.006	–
GRACE	0.008 ± 0.018	0.134 ± 0.121	0.181 ± 0.110	$\mathbf{0.001 \pm 0.001}$	0.001 ± 0.002	0.059 ± 0.072	0.422 ± 1.087	0.011 ± 0.001	0.011 ± 0.020	0.271 ± 0.073	–	–	0.005	–
TRACE	0.008 ± 0.028	0.103 ± 0.068	0.241 ± 0.248	0.001 ± 0.001	$\mathbf{0.000 \pm 0.000}$	0.017 ± 0.020	0.213 ± 0.094	0.140 ± 0.430	0.013 ± 0.010	0.232 ± 0.022	–	–	0.006	–
Framingham	0.003 ± 0.006	0.065 ± 0.048	0.073 ± 0.078	0.001 ± 0.001	$\mathbf{0.000 \pm 0.000}$	0.027 ± 0.036	0.380 ± 0.213	0.188 ± 0.399	0.019 ± 0.018	0.365 ± 0.077	–	–	0.009	–
Blood Transfusion	0.030 ± 0.058	0.225 ± 0.078	0.275 ± 0.200	0.009 ± 0.024	$\mathbf{0.000 \pm 0.001}$	0.089 ± 0.049	0.292 ± 0.265	0.026 ± 0.017	0.014 ± 0.017	0.632 ± 0.039	–	–	0.059	–
QSAR Biodeg	0.026 ± 0.028	0.110 ± 0.011	0.189 ± 0.041	0.149 ± 0.153	0.572 ± 0.052	0.034 ± 0.027	0.568 ± 0.387	0.025 ± 0.016	0.036 ± 0.056	0.680 ± 0.030	–	–	**0.015**	–
Maternal Risk	$\mathbf{0.001 \pm 0.000}$	0.141 ± 0.085	0.230 ± 0.151	0.015 ± 0.011	0.151 ± 0.183	0.066 ± 0.165	0.316 ± 0.525	0.019 ± 0.023	0.017 ± 0.019	0.360 ± 0.064	0.005	–	0.015	0.047
Website Phishing	0.043 ± 0.155	0.093 ± 0.001	0.138 ± 0.018	0.091 ± 0.002	$\mathbf{0.000 \pm 0.000}$	0.005 ± 0.009	0.083 ± 0.069	0.003 ± 0.005	0.009 ± 0.008	0.034 ± 0.006	–	–	0.003	–

Table 24: Fréchet-PCA (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	1.435 ± 1.803	4.871 ± 3.551	10.777 ± 1.890	$\mathbf{0.309 \pm 0.172}$	36.258	1.474 ± 0.435	11.078 ± 7.624	1.620 ± 0.978	0.651 ± 0.427	36.526 ± 6.670	1.810	0.357	0.870	1.287
Breast Cancer	9.030 ± 1.752	29.409 ± 5.766	13.849 ± 4.611	$\mathbf{0.790 \pm 0.154}$	262.006	5.478 ± 2.510	107.847 ± 59.046	12.971 ± 1.348	11.090 ± 12.826	216.031 ± 50.704	–	–	7.082	–
Liver Disorders	$\mathbf{0.083 \pm 0.095}$	5.592 ± 4.179	2.911 ± 0.827	0.102 ± 0.040	0.163	1.003 ± 1.006	3.041 ± 2.091	1.369 ± 0.855	0.689 ± 0.564	26.046 ± 2.381	1.348	–	0.794	–
Parkinsons	4.683 ± 0.535	20.827 ± 6.418	12.444 ± 2.154	$\mathbf{0.514 \pm 0.286}$	77.241	7.191 ± 3.318	19.322 ± 15.333	12.159 ± 1.849	5.990 ± 2.586	83.740 ± 5.762	–	–	7.445	–
Obesity	$\mathbf{0.150 \pm 0.160}$	1.640 ± 0.992	2.601 ± 0.085	1.213 ± 1.935	2.692	0.349 ± 0.768	4.702 ± 5.667	0.368 ± 0.063	0.948 ± 0.500	10.496 ± 2.717	–	–	0.184	–
Diabetes	1.132 ± 2.054	1.105 ± 0.353	0.953 ± 0.723	0.542 ± 0.953	–	0.263 ± 0.077	3.167 ± 1.559	0.796 ± 1.901	0.996 ± 1.917	27.853 ± 17.394	1.468	–	**0.079**	3.379
German Credit	3.048 ± 6.584	1.687 ± 1.583	8.982 ± 0.810	0.254 ± 0.129	–	0.435 ± 0.334	2.723 ± 1.713	0.581 ± 0.146	1.105 ± 0.711	16.535 ± 1.822	–	–	**0.224**	–
Bank Marketing	0.216 ± 0.195	0.585 ± 0.580	2.557 ± 1.284	3.818 ± 12.599	–	1.044 ± 0.846	12.075 ± 30.720	0.902 ± 0.377	8.494 ± 29.701	1835.097 ± 5924.108	3.189	–	**0.182**	–
Census Income	0.804 ± 0.891	0.799 ± 0.778	3.091 ± 1.218	21.677 ± 39.564	–	1.784 ± 1.003	7.884 ± 15.023	2.709 ± 1.327	1.540 ± 1.072	113.086 ± 33.070	23.296	–	**0.484**	–
Melanoma	0.392 ± 1.034	3.143 ± 2.444	3.381 ± 1.659	0.352 ± 0.782	0.543 ± 0.510	0.801 ± 1.452	4.625 ± 4.366	0.859 ± 0.984	0.386 ± 0.476	4.857 ± 2.184	**0.271**	–	0.607	–
PBC	1.455 ± 2.439	0.685 ± 0.733	3.424 ± 0.507	0.234 ± 0.485	$\mathbf{0.062 \pm 0.049}$	0.475 ± 0.287	1.865 ± 0.658	2.770 ± 9.431	0.439 ± 0.332	4.204 ± 0.987	0.192	–	0.363	–
PBC3	1.037 ± 1.425	3.377 ± 2.267	6.796 ± 1.308	$\mathbf{0.410 \pm 0.261}$	21.966 ± 14.353	1.710 ± 1.121	11.014 ± 5.263	2.785 ± 0.549	1.009 ± 0.265	28.665 ± 2.528	1.802	–	1.325	2.058
Retinopathy	0.924 ± 2.539	1.901 ± 1.251	3.689 ± 0.839	1.010 ± 0.404	$\mathbf{0.071 \pm 0.089}$	0.824 ± 0.867	2.196 ± 1.095	0.966 ± 0.197	1.526 ± 0.668	2.611 ± 0.508	0.093	–	0.541	–
WHAS500	1.048 ± 1.801	7.186 ± 5.019	10.013 ± 2.041	$\mathbf{0.309 \pm 0.074}$	10.632 ± 13.606	0.955 ± 0.983	16.775 ± 10.154	1.925 ± 1.097	0.928 ± 1.290	67.247 ± 23.059	–	–	1.027	0.984
GBSG2	0.387 ± 0.777	2.908 ± 2.172	5.838 ± 0.697	0.546 ± 0.458	2.622 ± 3.262	2.080 ± 3.255	8.209 ± 14.802	1.058 ± 0.462	0.470 ± 0.379	62.397 ± 16.244	0.931	–	**0.165**	–
Colon	0.401 ± 0.396	1.440 ± 1.317	6.550 ± 1.731	0.199 ± 0.096	$\mathbf{0.136 \pm 0.118}$	1.398 ± 2.134	5.128 ± 3.767	0.730 ± 0.497	0.493 ± 0.805	26.894 ± 8.398	2.278	–	0.300	–
GRACE	0.113 ± 0.192	2.794 ± 2.613	2.638 ± 1.011	$\mathbf{0.071 \pm 0.021}$	0.183 ± 0.247	0.998 ± 0.921	9.721 ± 23.582	0.345 ± 0.082	0.194 ± 0.296	11.076 ± 5.627	–	–	0.131	–
TRACE	0.299 ± 1.026	2.583 ± 1.281	5.195 ± 2.293	0.072 ± 0.042	$\mathbf{0.012 \pm 0.007}$	0.659 ± 0.398	3.817 ± 2.281	3.200 ± 8.118	0.410 ± 0.376	8.624 ± 0.478	–	–	0.185	–
Framingham	0.077 ± 0.106	2.388 ± 1.180	1.777 ± 1.239	$\mathbf{0.021 \pm 0.020}$	0.125 ± 0.254	0.623 ± 0.708	10.909 ± 10.687	4.702 ± 9.962	0.590 ± 0.403	29.902 ± 8.845	–	–	0.247	–
Blood Transfusion	0.938 ± 1.399	3.815 ± 0.776	3.424 ± 1.300	0.236 ± 0.581	$\mathbf{0.178 \pm 0.298}$	1.406 ± 0.086	3.423 ± 2.467	1.012 ± 0.429	0.921 ± 0.542	38.142 ± 4.669	–	–	0.864	–
QSAR Biodeg	7.542 ± 4.440	17.631 ± 3.741	20.915 ± 2.773	22.391 ± 27.121	1013.505 ± 203.490	4.580 ± 1.868	59.061 ± 54.876	9.061 ± 5.276	10.922 ± 9.358	754.443 ± 102.561	–	–	**4.203**	–
Maternal Risk	$\mathbf{0.112 \pm 0.094}$	4.425 ± 2.743	4.762 ± 0.952	0.675 ± 0.925	16.795 ± 19.484	1.322 ± 3.054	5.393 ± 6.872	0.583 ± 0.543	0.632 ± 0.218	18.273 ± 4.293	0.372	–	0.476	1.096
Website Phishing	0.562 ± 1.887	1.231 ± 0.144	2.260 ± 0.182	1.212 ± 0.084	$\mathbf{0.024 \pm 0.019}$	0.119 ± 0.090	1.272 ± 0.917	0.099 ± 0.087	0.198 ± 0.118	0.612 ± 0.098	–	–	0.089	–

Table 25: Wasserstein (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.225 ± 0.228	0.391 ± 0.066	0.597 ± 0.056	0.111 ± 0.027	0.958	0.247 ± 0.038	0.685 ± 0.228	0.308 ± 0.058	0.142 ± 0.063	1.085 ± 0.106	0.176	**0.094**	0.182	0.188
Breast Cancer	$\mathbf{0.143 \pm 0.160}$	0.692 ± 0.122	0.393 ± 0.103	0.150 ± 0.026	1.851	0.309 ± 0.043	1.567 ± 0.561	0.510 ± 0.065	0.167 ± 0.172	2.098 ± 0.235	–	–	0.322	–
Liver Disorders	$\mathbf{0.086 \pm 0.077}$	0.863 ± 0.148	0.423 ± 0.139	0.109 ± 0.024	0.113	0.356 ± 0.066	0.486 ± 0.236	0.425 ± 0.089	0.236 ± 0.149	1.872 ± 0.095	0.247	–	0.356	–
Parkinsons	0.121 ± 0.170	0.622 ± 0.122	0.406 ± 0.061	$\mathbf{0.120 \pm 0.014}$	1.227	0.384 ± 0.126	0.471 ± 0.482	0.540 ± 0.058	0.164 ± 0.045	1.477 ± 0.048	–	–	0.396	–
Obesity	$\mathbf{0.070 \pm 0.046}$	0.308 ± 0.151	0.358 ± 0.054	0.296 ± 0.123	0.300	0.172 ± 0.133	0.539 ± 0.618	0.175 ± 0.016	0.187 ± 0.091	0.727 ± 0.069	–	–	0.119	–
Diabetes	0.130 ± 0.149	0.245 ± 0.016	0.164 ± 0.063	0.184 ± 0.096	–	0.144 ± 0.036	0.471 ± 0.165	0.185 ± 0.196	0.207 ± 0.098	1.483 ± 0.399	0.230	–	**0.084**	0.508
German Credit	0.166 ± 0.046	0.252 ± 0.099	0.663 ± 0.091	$\mathbf{0.077 \pm 0.028}$	–	0.157 ± 0.070	0.406 ± 0.157	0.185 ± 0.024	0.190 ± 0.115	1.113 ± 0.097	–	–	0.091	–
Bank Marketing	$\mathbf{0.067 \pm 0.062}$	0.133 ± 0.066	0.245 ± 0.048	0.272 ± 0.184	–	0.184 ± 0.067	0.648 ± 0.882	0.201 ± 0.019	0.221 ± 0.179	7.511 ± 8.825	0.324	–	0.090	–
Census Income	0.132 ± 0.113	0.141 ± 0.084	0.264 ± 0.113	0.762 ± 1.192	–	0.248 ± 0.047	0.623 ± 0.925	0.202 ± 0.050	0.194 ± 0.096	3.209 ± 0.490	0.364	–	**0.077**	–
Melanoma	0.197 ± 0.292	0.580 ± 0.235	0.547 ± 0.213	$\mathbf{0.132 \pm 0.109}$	0.176 ± 0.078	0.273 ± 0.278	0.726 ± 0.538	0.333 ± 0.108	0.183 ± 0.068	0.686 ± 0.139	0.165	–	0.139	–
PBC	0.217 ± 0.111	0.195 ± 0.066	0.406 ± 0.095	0.090 ± 0.126	$\mathbf{0.055 \pm 0.026}$	0.226 ± 0.064	0.417 ± 0.154	0.354 ± 0.853	0.215 ± 0.095	0.510 ± 0.041	0.108	–	0.122	–
PBC3	0.192 ± 0.233	0.380 ± 0.131	0.422 ± 0.057	$\mathbf{0.098 \pm 0.043}$	0.786 ± 0.324	0.290 ± 0.122	0.718 ± 0.237	0.385 ± 0.055	0.168 ± 0.022	1.194 ± 0.035	0.214	–	0.208	0.273
Retinopathy	0.104 ± 0.090	0.191 ± 0.259	0.622 ± 0.134	0.109 ± 0.070	$\mathbf{0.052 \pm 0.017}$	0.204 ± 0.179	0.305 ± 0.222	0.193 ± 0.075	0.215 ± 0.153	0.323 ± 0.024	0.091	–	0.175	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
WHAS500	0.151 ± 0.209	0.435 ± 0.195	0.491 ± 0.065	**0.070 ± 0.016**	0.312 ± 0.171	0.176 ± 0.083	0.615 ± 0.359	0.298 ± 0.126	0.127 ± 0.031	1.128 ± 0.056	–	–	0.154	0.135
GBSG2	**0.044 ± 0.003**	0.524 ± 0.172	0.547 ± 0.110	0.172 ± 0.069	0.231 ± 0.153	0.405 ± 0.369	0.801 ± 0.599	0.321 ± 0.052	0.124 ± 0.032	2.844 ± 0.265	0.242	–	0.147	–
Colon	0.119 ± 0.076	0.219 ± 0.080	0.448 ± 0.125	0.072 ± 0.046	**0.050 ± 0.012**	0.268 ± 0.234	0.501 ± 0.337	0.166 ± 0.051	0.117 ± 0.081	1.267 ± 0.162	0.199	–	0.119	–
GRACE	0.114 ± 0.081	0.477 ± 0.248	0.521 ± 0.116	**0.075 ± 0.026**	0.100 ± 0.043	0.336 ± 0.199	0.967 ± 1.322	0.247 ± 0.036	0.153 ± 0.104	0.941 ± 0.126	–	–	0.122	–
TRACE	0.090 ± 0.149	0.414 ± 0.163	0.487 ± 0.212	0.048 ± 0.034	**0.032 ± 0.007**	0.209 ± 0.034	0.547 ± 0.186	0.432 ± 0.568	0.175 ± 0.088	0.885 ± 0.017	–	–	0.097	–
Framingham	0.080 ± 0.074	0.398 ± 0.211	0.280 ± 0.059	0.056 ± 0.024	**0.037 ± 0.023**	0.232 ± 0.127	0.985 ± 0.375	0.581 ± 0.872	0.214 ± 0.130	1.341 ± 0.154	–	–	0.116	–
Blood Transfusion	0.133 ± 0.153	0.636 ± 0.264	0.577 ± 0.137	0.189 ± 0.202	**0.117 ± 0.033**	0.459 ± 0.094	0.658 ± 0.258	0.323 ± 0.090	0.167 ± 0.116	2.406 ± 0.154	–	–	0.349	–
QSAR Biodeg	0.154 ± 0.101	0.398 ± 0.018	0.343 ± 0.027	0.364 ± 0.090	3.165 ± 0.165	0.192 ± 0.058	0.787 ± 0.514	0.244 ± 0.025	0.184 ± 0.084	3.328 ± 0.200	–	–	**0.148**	–
Maternal Risk	**0.084 ± 0.031**	0.615 ± 0.273	0.559 ± 0.072	0.255 ± 0.075	0.933 ± 0.560	0.323 ± 0.394	0.694 ± 0.565	0.304 ± 0.062	0.174 ± 0.060	1.182 ± 0.096	0.111	–	0.229	0.270
Website Phishing	–	–	–	–	–	–	–	–	–	–	–	–	–	–

Table 26: Association error (↓) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.067 ± 0.051	0.059 ± 0.005	0.070 ± 0.006	0.055 ± 0.007	0.150	0.069 ± 0.012	0.094 ± 0.022	0.060 ± 0.012	0.058 ± 0.008	0.063 ± 0.009	0.064	**0.046**	0.082	0.074
Breast Cancer	0.258 ± 0.014	0.347 ± 0.010	0.195 ± 0.007	**0.051 ± 0.017**	0.188	0.107 ± 0.028	0.290 ± 0.050	0.240 ± 0.022	0.298 ± 0.045	0.357 ± 0.012	–	–	0.236	–
Liver Disorders	0.086 ± 0.074	0.222 ± 0.028	0.156 ± 0.099	**0.052 ± 0.005**	0.080	0.118 ± 0.122	0.216 ± 0.077	0.207 ± 0.025	0.171 ± 0.019	0.232 ± 0.048	0.106	–	0.103	–
Parkinsons	0.234 ± 0.041	0.444 ± 0.026	0.335 ± 0.008	**0.056 ± 0.018**	0.224	0.175 ± 0.029	0.366 ± 0.025	0.369 ± 0.044	0.378 ± 0.054	0.450 ± 0.016	–	–	0.141	–
Obesity	0.044 ± 0.003	0.103 ± 0.004	0.086 ± 0.008	0.069 ± 0.006	0.115	**0.039 ± 0.058**	0.091 ± 0.017	0.074 ± 0.008	0.087 ± 0.013	0.105 ± 0.002	–	–	0.040	–
Diabetes	0.056 ± 0.044	0.090 ± 0.048	0.056 ± 0.028	0.071 ± 0.014	–	**0.027 ± 0.003**	0.079 ± 0.027	0.075 ± 0.018	0.065 ± 0.010	0.098 ± 0.006	0.054	–	0.048	0.151
German Credit	0.184 ± 0.342	0.085 ± 0.004	0.111 ± 0.014	0.072 ± 0.006	–	**0.048 ± 0.010**	0.082 ± 0.003	0.063 ± 0.013	0.080 ± 0.004	0.086 ± 0.003	–	–	0.064	–
Bank Marketing	**0.033 ± 0.007**	0.075 ± 0.021	0.079 ± 0.023	0.067 ± 0.004	–	0.034 ± 0.014	0.084 ± 0.009	0.043 ± 0.023	0.072 ± 0.005	0.089 ± 0.003	0.057	–	0.048	–
Census Income	0.050 ± 0.017	0.081 ± 0.007	0.075 ± 0.013	0.108 ± 0.017	–	**0.044 ± 0.011**	0.116 ± 0.015	0.070 ± 0.036	0.104 ± 0.007	0.127 ± 0.002	0.102	–	0.064	–
Melanoma	0.088 ± 0.017	0.127 ± 0.027	0.108 ± 0.033	0.097 ± 0.053	0.096 ± 0.023	0.086 ± 0.068	0.119 ± 0.076	0.113 ± 0.023	0.109 ± 0.086	0.130 ± 0.008	**0.039**	–	0.122	–
PBC	0.104 ± 0.113	0.089 ± 0.021	0.137 ± 0.042	0.078 ± 0.033	**0.053 ± 0.016**	0.060 ± 0.045	0.087 ± 0.016	0.082 ± 0.046	0.073 ± 0.026	0.090 ± 0.037	0.082	–	0.104	–
PBC3	0.080 ± 0.033	0.101 ± 0.018	0.104 ± 0.019	0.071 ± 0.017	0.101 ± 0.036	**0.068 ± 0.034**	0.099 ± 0.013	0.086 ± 0.009	0.096 ± 0.002	0.097 ± 0.010	0.106	–	0.091	0.095
Retinopathy	0.082 ± 0.091	0.118 ± 0.024	0.208 ± 0.145	0.092 ± 0.031	0.041 ± 0.036	0.088 ± 0.029	0.120 ± 0.010	0.093 ± 0.022	0.097 ± 0.038	0.120 ± 0.007	**0.037**	–	0.086	–
WHAS500	0.066 ± 0.025	0.074 ± 0.005	0.089 ± 0.007	**0.054 ± 0.007**	0.088 ± 0.031	0.055 ± 0.015	0.084 ± 0.029	0.064 ± 0.005	0.066 ± 0.009	0.071 ± 0.004	–	–	0.084	0.085
GBSG2	0.061 ± 0.030	0.124 ± 0.017	0.104 ± 0.018	0.077 ± 0.019	0.078 ± 0.063	0.098 ± 0.088	0.126 ± 0.008	0.090 ± 0.043	0.092 ± 0.036	0.121 ± 0.030	0.091	–	**0.050**	–
Colon	0.034 ± 0.010	0.053 ± 0.002	0.081 ± 0.015	0.044 ± 0.011	0.036 ± 0.005	0.082 ± 0.055	0.061 ± 0.013	0.043 ± 0.013	0.049 ± 0.015	0.057 ± 0.003	**0.033**	–	0.049	–
GRACE	**0.033 ± 0.016**	0.061 ± 0.017	0.108 ± 0.063	0.037 ± 0.004	0.048 ± 0.056	0.044 ± 0.018	0.088 ± 0.060	0.060 ± 0.023	0.040 ± 0.039	0.055 ± 0.010	–	–	0.088	–
TRACE	0.029 ± 0.017	0.102 ± 0.008	0.077 ± 0.022	0.064 ± 0.007	**0.017 ± 0.006**	0.062 ± 0.053	0.095 ± 0.030	0.104 ± 0.045	0.066 ± 0.040	0.109 ± 0.007	–	–	0.069	–
Framingham	0.020 ± 0.009	0.136 ± 0.022	0.259 ± 0.052	**0.014 ± 0.006**	0.029 ± 0.021	0.032 ± 0.005	0.144 ± 0.045	0.107 ± 0.003	0.113 ± 0.020	0.152 ± 0.008	–	–	0.097	–
Blood Transfusion	0.208 ± 0.547	0.412 ± 0.027	0.250 ± 0.080	0.069 ± 0.038	**0.031 ± 0.051**	0.120 ± 0.110	0.369 ± 0.086	0.315 ± 0.176	0.266 ± 0.145	0.402 ± 0.057	–	–	0.098	–
QSAR Biodeg	0.131 ± 0.013	0.147 ± 0.004	0.128 ± 0.018	**0.079 ± 0.009**	0.131 ± 0.012	0.083 ± 0.009	0.147 ± 0.019	0.117 ± 0.001	0.142 ± 0.008	0.147 ± 0.003	–	–	0.109	–
Maternal Risk	**0.070 ± 0.044**	0.194 ± 0.022	0.142 ± 0.033	0.071 ± 0.017	0.158 ± 0.077	0.081 ± 0.037	0.166 ± 0.044	0.114 ± 0.044	0.160 ± 0.034	0.205 ± 0.010	0.140	–	0.112	0.204
Website Phishing	0.081 ± 0.082	0.100 ± 0.009	0.187 ± 0.072	0.091 ± 0.012	**0.030 ± 0.006**	0.058 ± 0.013	0.105 ± 0.005	0.076 ± 0.012	0.067 ± 0.014	0.106 ± 0.008	–	–	0.058	–

SDMetrics quality report

Table 27: Overall quality (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.402 ± 0.050	0.369 ± 0.021	0.307 ± 0.016	0.423 ± 0.010	0.341	0.387 ± 0.010	0.297 ± 0.060	0.348 ± 0.017	0.415 ± 0.021	0.317 ± 0.015	0.407	**0.438**	0.402	0.402
Breast Cancer	0.770 ± 0.043	0.600 ± 0.005	0.719 ± 0.033	**0.849 ± 0.004**	0.704	0.783 ± 0.017	0.404 ± 0.149	0.614 ± 0.031	0.747 ± 0.003	0.522 ± 0.009	–	–	0.733	–
Liver Disorders	**0.873 ± 0.022**	0.597 ± 0.056	0.705 ± 0.040	0.870 ± 0.003	0.866	0.787 ± 0.024	0.635 ± 0.104	0.646 ± 0.040	0.760 ± 0.052	0.564 ± 0.021	0.832	–	0.767	–
Parkinsons	0.810 ± 0.059	0.556 ± 0.036	0.669 ± 0.012	**0.847 ± 0.012**	0.726	0.731 ± 0.059	0.603 ± 0.128	0.564 ± 0.020	0.724 ± 0.019	0.520 ± 0.018	–	–	0.744	–
Obesity	**0.906 ± 0.052**	0.775 ± 0.035	0.747 ± 0.050	0.816 ± 0.008	0.850	0.836 ± 0.053	0.718 ± 0.200	0.695 ± 0.033	0.792 ± 0.019	0.578 ± 0.006	–	–	0.800	–
Diabetes	**0.414 ± 0.015**	0.391 ± 0.005	0.399 ± 0.009	0.404 ± 0.014	–	0.396 ± 0.016	0.336 ± 0.060	0.388 ± 0.022	0.388 ± 0.025	0.272 ± 0.011	0.374	–	0.411	0.357
German Credit	0.610 ± 0.590	0.761 ± 0.019	0.557 ± 0.073	0.821 ± 0.007	–	0.816 ± 0.015	0.732 ± 0.022	0.785 ± 0.023	0.750 ± 0.014	0.542 ± 0.010	–	–	**0.828**	–
Bank Marketing	0.739 ± 0.007	0.709 ± 0.017	0.690 ± 0.024	0.702 ± 0.014	–	0.648 ± 0.100	0.511 ± 0.111	0.586 ± 0.116	0.614 ± 0.053	0.404 ± 0.004	0.631	–	**0.761**	–
Census Income	0.594 ± 0.092	0.590 ± 0.026	0.607 ± 0.044	0.597 ± 0.060	–	0.541 ± 0.089	0.402 ± 0.289	0.531 ± 0.106	0.502 ± 0.049	0.305 ± 0.002	0.523	–	**0.628**	–
Melanoma	0.366 ± 0.075	0.310 ± 0.003	0.288 ± 0.053	**0.387 ± 0.007**	0.384 ± 0.005	0.352 ± 0.049	0.285 ± 0.119	0.317 ± 0.033	0.376 ± 0.007	0.318 ± 0.006	0.380	–	0.377	–
PBC	0.456 ± 0.147	0.476 ± 0.192	0.450 ± 0.236	0.519 ± 0.276	**0.527 ± 0.270**	0.472 ± 0.224	0.445 ± 0.228	0.390 ± 0.086	0.500 ± 0.280	0.404 ± 0.150	0.455	–	0.506	–
PBC3	0.445 ± 0.165	0.416 ± 0.182	0.320 ± 0.020	0.480 ± 0.219	0.416 ± 0.226	0.434 ± 0.227	0.354 ± 0.198	0.378 ± 0.205	0.452 ± 0.198	0.344 ± 0.158	0.545	–	**0.560**	0.524
Retinopathy	0.777 ± 0.298	0.649 ± 0.071	0.611 ± 0.103	0.735 ± 0.038	0.881 ± 0.015	0.763 ± 0.106	0.606 ± 0.073	0.618 ± 0.016	0.720 ± 0.021	0.623 ± 0.030	**0.901**	–	0.814	–
WHAS500	0.534 ± 0.066	0.462 ± 0.069	0.451 ± 0.021	**0.560 ± 0.035**	0.524 ± 0.043	0.534 ± 0.018	0.424 ± 0.078	0.467 ± 0.061	0.530 ± 0.044	0.419 ± 0.030	–	–	0.494	0.523
GBSG2	0.755 ± 0.179	0.613 ± 0.097	0.472 ± 0.049	0.717 ± 0.140	**0.761 ± 0.161**	0.605 ± 0.208	0.562 ± 0.115	0.457 ± 0.120	0.720 ± 0.110	0.437 ± 0.135	0.669	–	0.680	–
Colon	0.882 ± 0.079	0.728 ± 0.030	0.696 ± 0.080	0.862 ± 0.013	**0.911 ± 0.027**	0.864 ± 0.065	0.669 ± 0.094	0.842 ± 0.007	0.827 ± 0.105	0.634 ± 0.037	0.874	–	0.895	–
GRACE	0.389 ± 0.015	0.337 ± 0.021	0.291 ± 0.024	**0.400 ± 0.005**	0.398 ± 0.006	0.340 ± 0.048	0.265 ± 0.200	0.350 ± 0.012	0.379 ± 0.026	0.316 ± 0.010	–	–	0.389	–
TRACE	0.412 ± 0.024	0.368 ± 0.028	0.331 ± 0.048	0.418 ± 0.005	**0.421 ± 0.001**	0.381 ± 0.018	0.330 ± 0.025	0.347 ± 0.099	0.392 ± 0.016	0.307 ± 0.006	–	–	0.403	–
Framingham	0.907 ± 0.014	0.678 ± 0.041	0.855 ± 0.016	0.920 ± 0.003	**0.927 ± 0.004**	0.867 ± 0.060	0.547 ± 0.021	0.692 ± 0.174	0.751 ± 0.053	0.600 ± 0.025	–	–	0.892	–
Blood Transfusion	0.673 ± 0.210	0.512 ± 0.070	0.547 ± 0.053	0.713 ± 0.014	**0.736 ± 0.015**	0.608 ± 0.043	0.474 ± 0.129	0.561 ± 0.056	0.643 ± 0.065	0.427 ± 0.026	–	–	0.640	–
QSAR Biodeg	0.761 ± 0.035	0.653 ± 0.014	0.686 ± 0.009	0.754 ± 0.026	0.576 ± 0.011	**0.768 ± 0.010**	0.550 ± 0.116	0.694 ± 0.024	0.712 ± 0.028	0.475 ± 0.013	–	–	0.764	–
Maternal Risk	0.824 ± 0.128	0.596 ± 0.043	0.505 ± 0.039	0.722 ± 0.028	0.657 ± 0.100	0.689 ± 0.112	0.539 ± 0.194	0.573 ± 0.005	0.782 ± 0.021	0.537 ± 0.045	**0.840**	–	0.711	0.724
Website Phishing	0.852 ± 0.243	0.765 ± 0.022	0.713 ± 0.012	0.796 ± 0.021	**0.965 ± 0.016**	0.910 ± 0.025	0.692 ± 0.099	0.902 ± 0.031	0.886 ± 0.005	0.750 ± 0.009	–	–	0.924	–

Table 28: Column shapes (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	0.804 ± 0.101	0.738 ± 0.042	0.614 ± 0.032	0.846 ± 0.020	0.681	0.773 ± 0.019	0.595 ± 0.120	0.696 ± 0.033	0.830 ± 0.041	0.634 ± 0.029	0.815	**0.875**	0.803	0.804
Breast Cancer	0.868 ± 0.077	0.649 ± 0.016	0.724 ± 0.082	0.868 ± 0.015	0.631	0.782 ± 0.028	0.255 ± 0.287	0.550 ± 0.060	**0.882 ± 0.039**	0.495 ± 0.027	–	–	0.675	–
Liver Disorders	**0.788 ± 0.032**	0.487 ± 0.106	0.612 ± 0.083	0.771 ± 0.011	0.772	0.632 ± 0.018	0.601 ± 0.179	0.535 ± 0.082	0.731 ± 0.068	0.434 ± 0.052	0.725	–	0.590	–
Parkinsons	0.861 ± 0.099	0.591 ± 0.048	0.700 ± 0.027	**0.863 ± 0.015**	0.676	0.714 ± 0.066	0.669 ± 0.236	0.522 ± 0.048	0.849 ± 0.016	0.522 ± 0.035	–	–	0.690	–

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Obesity	**0.941 ± 0.029**	0.856 ± 0.045	0.821 ± 0.027	0.893 ± 0.011	0.889	0.885 ± 0.003	0.794 ± 0.185	0.848 ± 0.024	0.879 ± 0.007	0.664 ± 0.013	–	–	0.891	–
Diabetes	**0.828 ± 0.030**	0.782 ± 0.009	0.798 ± 0.017	0.808 ± 0.028	–	0.791 ± 0.032	0.672 ± 0.119	0.776 ± 0.044	0.776 ± 0.050	0.543 ± 0.022	0.748	–	0.822	0.714
German Credit	0.690 ± 0.502	0.872 ± 0.017	0.639 ± 0.011	**0.925 ± 0.007**	–	0.885 ± 0.019	0.817 ± 0.061	0.870 ± 0.016	0.854 ± 0.020	0.622 ± 0.009	–	–	0.899	–
Bank Marketing	0.878 ± 0.005	0.852 ± 0.019	0.812 ± 0.023	0.852 ± 0.021	–	0.809 ± 0.085	0.667 ± 0.139	0.767 ± 0.041	0.783 ± 0.052	0.515 ± 0.007	0.775	–	**0.893**	–
Census Income	0.809 ± 0.086	0.796 ± 0.057	0.782 ± 0.031	**0.818 ± 0.109**	–	0.726 ± 0.076	0.572 ± 0.403	0.730 ± 0.102	0.710 ± 0.102	0.426 ± 0.008	0.755	–	0.805	–
Melanoma	0.733 ± 0.150	0.620 ± 0.005	0.576 ± 0.107	**0.773 ± 0.014**	0.767 ± 0.009	0.703 ± 0.097	0.569 ± 0.237	0.634 ± 0.066	0.752 ± 0.013	0.637 ± 0.013	0.759	–	0.754	–
PBC	0.780 ± 0.037	0.790 ± 0.017	0.704 ± 0.032	0.823 ± 0.038	**0.837 ± 0.012**	0.767 ± 0.038	0.712 ± 0.053	0.744 ± 0.213	0.777 ± 0.025	0.679 ± 0.016	0.807	–	0.803	–
PBC3	0.808 ± 0.104	0.736 ± 0.053	0.640 ± 0.041	**0.859 ± 0.014**	0.737 ± 0.052	0.766 ± 0.031	0.619 ± 0.190	0.661 ± 0.043	0.819 ± 0.028	0.613 ± 0.008	0.818	–	0.806	0.775
Retinopathy	0.807 ± 0.035	0.814 ± 0.034	0.626 ± 0.047	0.841 ± 0.016	**0.850 ± 0.004**	0.794 ± 0.063	0.747 ± 0.078	0.795 ± 0.017	0.794 ± 0.016	0.768 ± 0.001	0.842	–	0.824	–
WHAS500	0.864 ± 0.091	0.770 ± 0.068	0.712 ± 0.023	**0.904 ± 0.004**	0.855 ± 0.032	0.855 ± 0.039	0.725 ± 0.145	0.767 ± 0.057	0.880 ± 0.016	0.689 ± 0.016	–	–	0.853	0.869
GBSG2	**0.838 ± 0.033**	0.710 ± 0.072	0.566 ± 0.031	0.808 ± 0.003	0.831 ± 0.021	0.687 ± 0.205	0.635 ± 0.242	0.701 ± 0.024	0.824 ± 0.030	0.512 ± 0.023	0.747	–	0.798	–
Colon	0.785 ± 0.160	0.823 ± 0.015	0.619 ± 0.065	0.872 ± 0.011	**0.879 ± 0.005**	0.754 ± 0.118	0.699 ± 0.154	0.815 ± 0.022	0.838 ± 0.021	0.630 ± 0.017	0.755	–	0.834	–
GRACE	0.777 ± 0.031	0.675 ± 0.041	0.582 ± 0.047	**0.800 ± 0.010**	0.797 ± 0.013	0.679 ± 0.097	0.531 ± 0.400	0.700 ± 0.024	0.757 ± 0.053	0.632 ± 0.020	–	–	0.778	–
TRACE	0.824 ± 0.048	0.735 ± 0.056	0.663 ± 0.096	0.837 ± 0.010	**0.843 ± 0.003**	0.763 ± 0.035	0.660 ± 0.049	0.695 ± 0.198	0.783 ± 0.031	0.613 ± 0.011	–	–	0.807	–
Framingham	0.830 ± 0.024	0.723 ± 0.079	0.726 ± 0.031	0.844 ± 0.012	**0.860 ± 0.003**	0.759 ± 0.058	0.494 ± 0.018	0.615 ± 0.313	0.768 ± 0.049	0.599 ± 0.016	–	–	0.811	–
Blood Transfusion	0.693 ± 0.147	0.539 ± 0.148	0.483 ± 0.058	0.695 ± 0.031	**0.732 ± 0.017**	0.531 ± 0.006	0.434 ± 0.207	0.576 ± 0.070	0.694 ± 0.052	0.370 ± 0.023	–	–	0.582	–
QSAR Biodeg	**0.881 ± 0.041**	0.772 ± 0.022	0.777 ± 0.021	0.791 ± 0.030	0.561 ± 0.034	0.849 ± 0.029	0.574 ± 0.167	0.775 ± 0.019	0.873 ± 0.005	0.417 ± 0.015	–	–	0.874	–
Maternal Risk	0.931 ± 0.068	0.663 ± 0.019	0.630 ± 0.029	0.764 ± 0.038	0.730 ± 0.073	0.732 ± 0.233	0.588 ± 0.268	0.724 ± 0.031	0.867 ± 0.044	0.585 ± 0.011	**0.940**	–	0.749	0.781
Website Phishing	0.908 ± 0.097	0.766 ± 0.016	0.686 ± 0.018	0.782 ± 0.009	**0.976 ± 0.010**	0.936 ± 0.032	0.769 ± 0.094	0.953 ± 0.033	0.920 ± 0.012	0.841 ± 0.014	–	–	0.951	–

Table 29: Column-pair trends (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000**	**0.000**	**0.000**	**0.000**
Breast Cancer	0.672 ± 0.012	0.550 ± 0.009	0.714 ± 0.017	**0.830 ± 0.015**	0.777	0.784 ± 0.021	0.553 ± 0.016	0.678 ± 0.008	0.611 ± 0.039	0.549 ± 0.010	–	–	0.792	–
Liver Disorders	0.958 ± 0.035	0.707 ± 0.008	0.798 ± 0.098	**0.969 ± 0.006**	0.959	0.942 ± 0.060	0.668 ± 0.035	0.757 ± 0.004	0.790 ± 0.036	0.694 ± 0.018	0.940	–	0.944	–
Parkinsons	0.760 ± 0.044	0.520 ± 0.025	0.637 ± 0.017	**0.831 ± 0.009**	0.777	0.749 ± 0.052	0.537 ± 0.028	0.606 ± 0.030	0.599 ± 0.042	0.518 ± 0.008	–	–	0.799	–
Obesity	**0.870 ± 0.076**	0.695 ± 0.026	0.673 ± 0.074	0.738 ± 0.008	0.812	0.788 ± 0.104	0.642 ± 0.217	0.541 ± 0.070	0.705 ± 0.037	0.492 ± 0.010	–	–	0.709	–
Diabetes	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000**	–	**0.000**	**0.000**
German Credit	0.529 ± 0.680	0.650 ± 0.033	0.475 ± 0.151	0.716 ± 0.020	–	0.746 ± 0.011	0.647 ± 0.045	0.701 ± 0.059	0.646 ± 0.039	0.462 ± 0.015	–	–	**0.757**	–
Bank Marketing	0.601 ± 0.019	0.566 ± 0.016	0.568 ± 0.027	0.553 ± 0.014	–	0.488 ± 0.116	0.356 ± 0.117	0.406 ± 0.192	0.445 ± 0.054	0.293 ± 0.007	0.487	–	**0.630**	–
Census Income	0.379 ± 0.103	0.383 ± 0.025	0.431 ± 0.062	0.375 ± 0.027	–	0.355 ± 0.103	0.233 ± 0.177	0.333 ± 0.124	0.293 ± 0.011	0.185 ± 0.003	0.291	–	**0.451**	–
Melanoma	**-0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000**	–	**0.000**	–
PBC	0.132 ± 0.297	0.163 ± 0.392	0.197 ± 0.484	0.215 ± 0.544	0.217 ± 0.539	0.178 ± 0.449	0.178 ± 0.505	0.035 ± 0.076	**0.222 ± 0.559**	0.129 ± 0.306	0.102	–	0.210	–
PBC3	0.082 ± 0.353	0.096 ± 0.415	0.000 ± 0.000	0.101 ± 0.434	0.094 ± 0.404	0.102 ± 0.439	0.090 ± 0.386	0.096 ± 0.411	0.086 ± 0.368	0.075 ± 0.324	0.272	–	**0.314**	0.272
Retinopathy	0.746 ± 0.562	0.485 ± 0.108	0.595 ± 0.237	0.628 ± 0.087	0.911 ± 0.034	0.733 ± 0.153	0.465 ± 0.103	0.440 ± 0.024	0.646 ± 0.036	0.479 ± 0.062	**0.959**	–	0.804	–
WHAS500	0.204 ± 0.059	0.155 ± 0.070	0.190 ± 0.057	**0.216 ± 0.073**	0.192 ± 0.057	0.214 ± 0.069	0.123 ± 0.023	0.166 ± 0.067	0.181 ± 0.074	0.148 ± 0.072	–	–	0.135	0.177
GBSG2	0.673 ± 0.348	0.515 ± 0.144	0.378 ± 0.070	0.626 ± 0.278	**0.692 ± 0.339**	0.522 ± 0.234	0.489 ± 0.109	0.213 ± 0.264	0.616 ± 0.192	0.362 ± 0.256	0.591	–	0.562	–
Colon	0.979 ± 0.007	0.634 ± 0.056	0.772 ± 0.096	0.852 ± 0.036	0.943 ± 0.050	0.974 ± 0.064	0.639 ± 0.044	0.869 ± 0.017	0.816 ± 0.201	0.638 ± 0.083	**0.993**	–	0.957	–
GRACE	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	–	–	**0.000**	–
TRACE	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	**-0.000 ± 0.000**	**0.000 ± 0.000**	–	–	**0.000**	–
Framingham	0.984 ± 0.018	0.634 ± 0.033	0.985 ± 0.009	**0.996 ± 0.006**	0.995 ± 0.005	0.975 ± 0.071	0.600 ± 0.061	0.769 ± 0.040	0.734 ± 0.093	0.602 ± 0.040	–	–	0.973	–
Blood Transfusion	0.653 ± 0.348	0.484 ± 0.011	0.610 ± 0.054	0.731 ± 0.006	**0.740 ± 0.016**	0.685 ± 0.085	0.514 ± 0.076	0.547 ± 0.101	0.593 ± 0.129	0.484 ± 0.040	–	–	0.698	–
QSAR Biodeg	0.640 ± 0.035	0.534 ± 0.037	0.595 ± 0.020	**0.716 ± 0.065**	0.591 ± 0.040	0.688 ± 0.042	0.526 ± 0.088	0.613 ± 0.039	0.551 ± 0.051	0.532 ± 0.035	–	–	0.655	–
Maternal Risk	0.717 ± 0.188	0.528 ± 0.090	0.381 ± 0.051	0.681 ± 0.037	0.584 ± 0.133	0.645 ± 0.014	0.491 ± 0.137	0.422 ± 0.023	0.698 ± 0.068	0.490 ± 0.079	**0.740**	–	0.674	0.667
Website Phishing	0.796 ± 0.390	0.765 ± 0.035	0.739 ± 0.036	0.809 ± 0.037	**0.955 ± 0.021**	0.883 ± 0.021	0.616 ± 0.106	0.851 ± 0.031	0.851 ± 0.005	0.660 ± 0.021	–	–	0.896	–

Other

Table 30: Parse valid-rate (↑) — mean ± 95% CI over seeds

Dataset	DISCO-TAB	CTGAN	TVAE	GaussCop	TabDDPM	ADS-GAN	PATE-GAN	RTVAE	NFlow	Marginals	GReaT	RealTabF	TAEGAN	TAGAL
Heart Failure	**0.675 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Breast Cancer	**0.693 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Liver Disorders	**0.680 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Parkinsons	**0.655 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Obesity	**0.708 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Diabetes	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
German Credit	**0.537 ± 0.708**	–	–	–	–	–	–	–	–	–	–	–	–	–
Bank Marketing	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Census Income	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Melanoma	**0.659 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
PBC	**0.474 ± 0.873**	–	–	–	–	–	–	–	–	–	–	–	–	–
PBC3	**0.680 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Retinopathy	**0.682 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
WHAS500	**0.688 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
GBSG2	**0.696 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Colon	**0.700 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
GRACE	**0.702 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
TRACE	**0.708 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Framingham	**0.711 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Blood Transfusion	**0.529 ± 0.701**	–	–	–	–	–	–	–	–	–	–	–	–	–
QSAR Biodeg	**0.703 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Maternal Risk	**0.688 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–
Website Phishing	**0.698 ± 0.000**	–	–	–	–	–	–	–	–	–	–	–	–	–